Explicit convergent Calabi-Yau metric representation on Kummer K3 surfaces

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Allowed standard prerequisites (editorial).

- Classical author-cited research prerequisites. Complete original embedded bibliography retained; every source-local main construction, support argument and explicitly non-adjacent continuation retained.

1 Introduction

Yau highlighted the problem of constructing an explicit metric with holonomy group $SU(2)$ on a K3 surface [13, pp. 333–334]. His theorem [12] guarantees a unique Ricci-flat Kähler metric in each Kähler class. We address the construction problem for a small-parameter family on a fixed Kummer K3 surface by prescribing a convergent series for the Calabi–Yau potential. We construct the background Green operator and compose it with the quadratic Monge–Ampère term to determine every correction recursively. The main theorem proves that the sum yields the metric supplied by the Calabi–Yau theorem.

Let $T = \mathbb{C}^2/(\mathbb{Z} + \sqrt{-1}\mathbb{Z})^2$, and let $\pi : X \to T/\{\pm 1\}$ be the minimal resolution. Denote the exceptional curves by $E_1, \dots, E_{16}$, and the pullback of the flat orbifold Kähler class by κ. We use $d^c = \frac{\sqrt{-1}}{2}(\bar\partial - \partial)$, so that $dd^c = \sqrt{-1}\partial\bar\partial$. The form $dz_1 \wedge dz_2$ extends to a nowhere-vanishing holomorphic two-form Ω on X. Set $\mu = \frac{1}{4}\Omega \wedge \bar\Omega$, with $\int_X \mu = \kappa^2/2 = 1/2$. For the Eguchi–Hanson scale $a > 0$, define

$$\alpha_a = \kappa - \frac{\pi a^2}{2}\sum_{\nu=1}^{16}[E_\nu], \qquad c_a = 1 - 8\pi^2 a^4. \tag{1.1}$$

Here $[E_\nu]$ is the Poincaré dual class; hence $\alpha_a \cdot E_\nu = \pi a^2$ and $\alpha_a^2/2 = c_a \int_X \mu$.

The classical Kummer construction joins the flat quotient metric to Eguchi–Hanson metrics [3] near the sixteen exceptional curves. Its analytic realization is treated by Kobayashi [8] and, as a model gluing problem, by Donaldson [2]; see also Joyce [6]. We choose an explicit radial interpolation giving a smooth Kähler form $\omega_a^{\mathrm{mod}} \in \alpha_a$ for small a. It is exactly Eguchi–Hanson near the exceptional curves and exactly flat outside fixed neighborhoods. Its volume form $dV_a = (\omega_a^{\mathrm{mod}})^2/2 = (1-\beta_a)\mu$ has an explicit density, with β_a supported in the transition annuli. The normalized residual is

$$f_a := \frac{c_a \mu}{dV_a} - 1 = \frac{\beta_a - 8\pi^2 a^4}{1 - \beta_a}.$$

It has zero dV_a-mean and satisfies $\|f_a\|_{C^{0,\alpha}_{\mathrm{sc}}} \le C a^4$, where $0 < \alpha < 1$ is fixed and the scale-adapted Hölder norms are defined in Section 2.6.

The background proves that α_a is Kähler on its construction range. Yau's theorem then supplies a smooth potential v_a, normalized by $\int_X v_a\, dV_a = 0$, such that

$$\omega_{\mathrm{CY},a} = \omega_a^{\mathrm{mod}} + \sqrt{-1}\partial\bar\partial v_a > 0, \qquad \frac{\omega_{\mathrm{CY},a}^2}{2} = c_a\mu. \tag{1.2}$$

No estimate for v_a is assumed. The bounds for its recursive representation follow from the background and its linear inverse. Yau's theorem applies whenever α_a is Kähler; the smaller range established below concerns convergence of the particular expansion about ω_a^{mod}.

All potentials and sources are real-valued. On functions of zero dV_a-mean, let $A_a = -\Lambda_{\omega_a^{\mathrm{mod}}}\sqrt{-1}\partial\bar\partial$ and $G_a = A_a^{-1}$, normalized by $\int_X G_a f\, dV_a = 0$. The symmetric bilinear map $Q_a(u,v) = \sqrt{-1}\partial\bar\partial u \wedge \sqrt{-1}\partial\bar\partial v/(\omega_a^{\mathrm{mod}})^2$ has zero mean by Stokes' theorem. In complex dimension two the volume equation is exactly quadratic:

$$-A_a u + Q_a(u,u) = f_a, \qquad u = -G_a f_a + G_a Q_a(u,u).$$

Thus the same explicitly represented operator G_aQ_a determines every order of correction.

Theorem 1.1 (Explicit representation of the Calabi–Yau metric)**.** *There exists $a_0 > 0$ such that, for every $0 < a < a_0$, the mean-zero functions defined by*

$$\begin{aligned} U_{a,1} &= -G_a f_a, \\ U_{a,n} &= G_a\left(\sum_{j=1}^{n-1} Q_a(U_{a,j}, U_{a,n-j})\right), \qquad n \ge 2, \end{aligned} \tag{1.3}$$

form an absolutely convergent series in $\mathcal{X}_a = C^{2,\alpha}(X,\mathbb{R})/\mathbb{R}$, endowed with the norm $\|u\|_{\mathcal{X}_a} = \|(\omega_a^{\mathrm{mod}})^{-1}\sqrt{-1}\partial\bar\partial u\|_{C^{0,\alpha}_{\mathrm{sc}}}$. Its sum is the normalized Calabi–Yau potential v_a, and

$$v_a = \sum_{n=1}^{\infty} U_{a,n}, \qquad \omega_{\mathrm{CY},a} = \omega_a^{\mathrm{mod}} + \sqrt{-1}\partial\bar\partial\sum_{n=1}^{\infty} U_{a,n}. \tag{1.4}$$

In particular, $\|v_a\|_{\mathcal{X}_a} \le Ca^3$. For constants $C_1, C_2 > 0$, independent of a and n,

$$\|U_{a,n}\|_{\mathcal{X}_a} \le \mathrm{Cat}_{n-1}\, C_1^n C_2^{n-1} a^{2n+1}, \qquad \mathrm{Cat}_m = \frac{1}{m+1}\binom{2m}{m}.$$

The background, residual, and Green operator are specified in Sections 2–3 and Appendices A–B.

The convergence is global on the smooth surface X, including the exceptional curves, for each sufficiently small fixed a. For fixed a, the displayed norm is equivalent to the ordinary $C^{2,\alpha}$-norm modulo constants. The terms depend on a through f_a, G_a, Q_a; the theorem does not assert a power series in a with parameter-independent coefficients.

The linear construction cuts along sixteen fixed interfaces, separating a flat exterior from sixteen caps. Let $\mathcal{D}_a$ be the block Dirichlet inverse, $\mathcal{H}_a$ the harmonic extension, and $\mathcal{J}_a$ the jump of outward normal derivatives. With $\mathcal{N}_a = \mathcal{J}_a\mathcal{H}_a$,

$$G_a f = (I - \Pi_a)\left(\mathcal{D}_a f - \mathcal{H}_a\mathcal{N}_a^{\dagger}\mathcal{J}_a\mathcal{D}_a f\right),$$

where $\Pi_a v = \int_X v\, dV_a / \int_X dV_a$. The interface inverse acts on compatible data modulo the common constant. Periodic Fourier–Ewald kernels describe the exterior, and angular separation gives cap kernels through convergent Frobenius and Volterra series. High-mode Neumann series and finite-dimensional Schur complements determine the remaining inverses.

Each application of G_a is therefore prescribed by volume and boundary integrations, convergent series, and finite-dimensional linear algebra. Only the radial kernel construction uses exclusively one-dimensional integrations. This is the analytic meaning of explicitness here; numerical certification requires the additional bounds discussed in Section 6.

The estimates governing the nonlinear series are

$$\|G_a f\|_{\mathcal{X}_a} \le C a^{-1} \|f\|_{\mathcal{Y}_a}, \qquad \|Q_a(u,v)\|_{\mathcal{Y}_a} \le \tfrac{1}{2}\|u\|_{\mathcal{X}_a}\|v\|_{\mathcal{X}_a},$$

where $\mathcal{Y}_a = C^{0,\alpha}_{\mathrm{sc}}(X)$ and f has zero mean. They give $\|G_a f_a\|_{\mathcal{X}_a} = O(a^3)$ and $\|G_a Q_a\|_{\mathrm{bil}} = O(a^{-1})$, so the sufficient convergence condition $4\|G_a f_a\|_{\mathcal{X}_a}\|G_a Q_a\|_{\mathrm{bil}} < 1$ holds for small a. Proposition 5.1 then supplies a Catalan majorant. Positivity follows along the convergent path $\sum_{n\ge1} t^n U_{a,n}$, $0 \le t \le 1$, without a separate smallness restriction. Integration by parts identifies the sum with v_a, and smoothness follows from that identification. The Calabi–Yau theorem supplies the target; the background estimates establish convergence of this recursion.

Explicit K3 expansions near torus orbifold limits appear in Kachru–Tripathy–Zimet [7, Section 3.4] and Tripathy–Zimet [11, Section 3.2]. Equations (3.77), (3.80), and (3.84) of the latter give a coefficient recursion in which a fixed linear solution operator is applied to quadratic combinations of lower-order corrections; its integral equation (3.93) has specified kernels. The discussion following that equation proposes a contraction argument for rigorous justification. These recursions concern hyper-Kähler quotient data on the dual torus. Our recursion acts on scalar Kähler potentials on the resolved compact Kummer surface. Donaldson's scalar perturbation argument [2, Sections 3.1–3.2] uses a bounded linear inverse and the inverse function theorem to solve the quadratic Monge–Ampère equation. His operator Q is linear and takes values in real $(1,1)$-forms, whereas Q_a above is a scalar-valued symmetric bilinear map. His argument does not give the coefficient recursion (1.3) or the explicit global Green representation used here.

The distinctive construction in this paper is the explicit realization of the bilinear correction operator $\mathcal{B}_a = G_a \circ Q_a$, where $G_a = A_a^{-1}$ is the normalized Green operator of the background scalar Laplace operator. We first represent G_a globally on the resolved compact surface by periodic and radial kernels, interface matching, finite-dimensional Schur complements, and convergent Neumann series. Starting from $U_{a,1} = -G_a f_a$, the same operator $\mathcal{B}_a$ determines every higher-order coefficient of the scalar Kähler potential through (1.3), giving the full binary-tree expansion. Finally, the Green and quadratic estimates yield a Catalan majorant proving absolute convergence on the entire smooth surface, including the exceptional curves, and comparison identifies the sum with the normalized Calabi–Yau potential in the family (1.1). The same majorant also gives the nonlinear truncation bound in (6.1). The general mechanism of applying a linear inverse to a quadratic source has precedents; the contribution here is this explicit global Green-operator realization of the convergent scalar-potential expansion.

The representation also provides geometric input for hyper-Kähler rotation and special Lagrangian geometry on X. With $\widehat{\Omega}_a = \sqrt{c_a}\,\Omega$, the forms $(\omega_{\mathrm{CY},a}, \operatorname{Re}\widehat{\Omega}_a, \operatorname{Im}\widehat{\Omega}_a)$ determine the hyper-Kähler triple, and its complex structures are obtained pointwise from the explicitly represented metric. For either elliptic fibration induced by projection onto a factor of T, every smooth fibre becomes a special Lagrangian torus under equatorial twistor rotation. This special Lagrangian assertion follows from standard hyper-Kähler rotation. The present construction supplies a convergent metric representation in the original resolution charts, together with an analytic truncation bound in $C^{0,\alpha}_{\mathrm{sc}}$. It thus provides a concrete starting point for writing the holomorphic-disc equations and studying instanton corrections. Disc existence and counts, quantitative coordinate expansions, and singular completion of the SYZ mirror require further geometric and analytic arguments.

In a forthcoming paper, we will extend our Green-operator method to construct explicit Calabi–Yau metrics on Calabi–Yau threefolds, including Kummer threefolds [1, p. 5] and the Schoen threefold [10]. We will then use these exact metrics to investigate special Lagrangian torus fibrations on these threefolds.

Section 2 constructs the background and residual. Sections 3 and 4 give the Green representation and its estimates. Section 5 sums the recursion and identifies its limit with the Calabi–Yau potential. Section 6 discusses evaluation and truncation. Finally, Section 7 briefly records sufficient conditions for the same mechanism on elliptic K3 surfaces. The elliptic Green construction and the corresponding global estimates are not proved in this paper. The appendices provide the periodic and radial kernel formulas.

The SYZ picture suggests that the smooth Calabi–Yau metric is obtained by correcting the semi-flat Calabi–Yau metric with instanton corrections. In the elliptic K3 case, the semi-flat metric was constructed by Greene–Shapere–Vafa–Yau [4]. Gross and Wilson [5] glued Ooguri–Vafa metrics to this background and corrected the resulting approximate metric by solving the complex Monge–Ampère equation, as in the proof of the Calabi conjecture. The final correction is obtained by an existence argument rather than an explicit convergent representation.

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2 Kummer geometry and the explicit background

We fix the geometric normalizations and construct a radial Kähler background which is exactly Eguchi–Hanson near the sixteen exceptional curves and exactly flat outside fixed neighborhoods. Its normalized Calabi–Yau residual is $O(a^4)$.

2.1 The flat Kummer orbifold and its resolution. Let $T = \mathbb{C}^2/(\mathbb{Z} + \sqrt{-1}\mathbb{Z})^2$ be the square complex two-torus. In the coordinates $z_j = x_j + \sqrt{-1}y_j$, $j = 1, 2$, on its universal cover, set

$$\omega_0 = \frac{\sqrt{-1}}{2}\left(dz_1 \wedge d\bar{z}_1 + dz_2 \wedge d\bar{z}_2\right). \tag{2.1}$$

The holomorphic form $\Omega_0 = dz_1 \wedge dz_2$ satisfies

$$dV_0 := \frac{\omega_0^2}{2} = \frac{1}{4}\Omega_0 \wedge \bar{\Omega}_0, \qquad \int_T dV_0 = 1. \tag{2.2}$$

The involution $\iota(z) = -z$ has sixteen fixed points, $T[2] = (\frac{1}{2}\mathbb{Z}/\mathbb{Z})^4$. Their images $p_1, \dots, p_{16}$ in $Y = T/\{\pm 1\}$ are A_1-singularities, locally isomorphic to $\mathbb{C}^2/\{\pm 1\}$. Let $\pi : X \to Y$ be the minimal resolution, with exceptional curves $E_1, \dots, E_{16}$. These are mutually disjoint smooth rational curves. The pullback $\kappa \in H^{1,1}(X, \mathbb{R})$ of the flat orbifold class satisfies

$$E_\nu^2 = -2, \quad E_\nu \cdot E_\mu = 0 \ (\nu \neq \mu), \quad \kappa \cdot E_\nu = 0, \quad \frac{\kappa^2}{2} = \frac{1}{2}. \tag{2.3}$$

The invariant form Ω_0 descends to the regular part of Y and extends across its crepant resolution to a nowhere-vanishing holomorphic two-form Ω on X. Thus $\mu = \frac{1}{4}\Omega \wedge \bar{\Omega}$ has total volume $\int_X \mu = 1/2$, since $T \to Y$ has degree two.

2.2 The Kummer Kähler classes. For the Eguchi–Hanson scale $a > 0$, define

$$\alpha_a = \kappa - \frac{\pi a^2}{2}\sum_{\nu=1}^{16}[E_\nu]. \tag{2.4}$$

Here $[E_\nu]$ is the Poincaré dual class. The intersection identities give $\alpha_a \cdot E_\nu = \pi a^2$ and

$$\frac{\alpha_a^2}{2} = \frac{\kappa^2}{2} + \frac{\pi^2 a^4}{8} \sum_{\nu=1}^{16} E_\nu^2 = \frac{1}{2} - 4\pi^2 a^4. \tag{2.5}$$

Consequently, the target volume constant is $c_a = \alpha_a^2/(2\int_X \mu) = 1 - 8\pi^2 a^4$, and the Calabi–Yau equation is $[\omega] = \alpha_a$, $\omega^2/2 = c_a \mu$. The construction below gives a Kähler representative on an explicit parameter range; it does not determine the full intersection of this family with the Kähler cone.

2.3 The Eguchi–Hanson model. On the punctured cover of $\mathbb{C}^2/\{\pm 1\}$, put $\rho = |z_1|^2 + |z_2|^2$. In the normalization of (2.1), the Eguchi–Hanson potential [3] is

$$\Phi_{\mathrm{EH},a}(\rho) = \frac{1}{2}\left[\sqrt{\rho^2 + a^4} + a^2 \log \frac{\rho}{\sqrt{\rho^2 + a^4} + a^2}\right]. \tag{2.6}$$

Set $\omega_{\mathrm{EH},a} = \sqrt{-1}\partial\bar\partial \Phi_{\mathrm{EH},a}$. Its complex Hessian has tangential and radial eigenvalues

$$A_{\mathrm{EH},a} = \Phi'_{\mathrm{EH},a} = \frac{\sqrt{\rho^2 + a^4}}{2\rho}, \qquad B_{\mathrm{EH},a} = \Phi'_{\mathrm{EH},a} + \rho\Phi''_{\mathrm{EH},a} = \frac{\rho}{2\sqrt{\rho^2 + a^4}}.$$

Primes denote differentiation in ρ. Since $4A_{\mathrm{EH},a}B_{\mathrm{EH},a} = 1$,

$$\frac{\omega_{\mathrm{EH},a}^2}{2} = \mu. \tag{2.7}$$

For fixed $a > 0$, its expansion at $\rho = 0$ is

$$\Phi_{\mathrm{EH},a}(\rho) = \frac{a^2}{2}\log\rho + \frac{a^2}{2}\big(1 - \log(2a^2)\big) + \frac{\rho^2}{8a^2} + O(\rho^4/a^6).$$

In the resolution chart $s = z_1^2$, $\xi = z_2/z_1$, one has $\rho = |s|(1 + |\xi|^2)$ and $E = \{s = 0\}$. Subtracting the pluriharmonic term $\frac{a^2}{2}\log|s|$ gives a potential extending smoothly across E. Its restriction, up to an additive constant, is $\frac{a^2}{2}\log(1 + |\xi|^2)$; the Fubini–Study form $\sqrt{-1}\partial\bar\partial\log(1 + |\xi|^2)$ has integral 2π, so

$$\int_E \omega_{\mathrm{EH},a} = \pi a^2. \tag{2.8}$$

This fixes the coefficient in (2.4).

For $\rho > a^2$, the potential has the convergent expansion

$$\Phi_{\mathrm{EH},a}(\rho) = \frac{\rho}{2} + \sum_{j=1}^{\infty} c_j a^{4j} \rho^{1-2j}, \qquad c_j = (-1)^j \frac{(2j-3)!!}{2^{j+1} j!(2j-1)}. \tag{2.9}$$

Here $j \geq 1$ and $(-1)!! = 1$. The first terms are

$$\Phi_{\mathrm{EH},a}(\rho) = \frac{\rho}{2} - \frac{a^4}{4\rho} + \frac{a^8}{48\rho^3} - \frac{a^{12}}{160\rho^5} + \frac{5a^{16}}{1792\rho^7} + O(a^{20}\rho^{-9}).$$

The coefficient expansion is useful for evaluation, but the construction uses the exact potential (2.6).

2.4 The radial interpolation. Fix radii

$$0 < r_1 < r_2 < r_3 < r_4 < R < \frac{1}{8}. \tag{2.10}$$

The sixteen orbifold balls of radius R are pairwise disjoint. For an explicit cutoff, set $\vartheta(t) = e^{-1/t}$ for $t > 0$ and $\vartheta(t) = 0$ for $t \le 0$, and let $\chi(t) = \vartheta(t)/(\vartheta(t)+\vartheta(1-t))$. Then χ is smooth, vanishes on $(-\infty, 0]$, and equals one on $[1, \infty)$. Define

$$\psi(r) = \chi\left(\frac{r - r_1}{r_2 - r_1}\right)\chi\left(\frac{r_4 - r}{r_4 - r_3}\right).$$

Thus $0 \le \psi \le 1$, with $\psi = 0$ on $[0, r_1]\cup[r_4, \infty)$ and $\psi = 1$ on $[r_2, r_3]$. Put $I_\psi = \int_0^\infty r^3\psi(r)\,dr > 0$ and, with $\rho = r^2$, define

$$b_a = \frac{a^4}{4I_\psi}, \qquad \beta_a(\rho) = b_a\psi(\sqrt{\rho}), \qquad m_a(\rho) = \int_0^\rho s\beta_a(s)\,ds. \tag{2.11}$$

Changing variables gives $m_a(\infty) = 2b_aI_\psi = a^4/2$, hence $0 \le m_a \le a^4/2$. Set

$$\begin{aligned} &D_a(\rho) = \rho^2 + a^4 - 2m_a(\rho) \ge \rho^2, && A_a^{\mathrm{mod}} = \frac{\sqrt{D_a}}{2\rho}, \\ &B_a^{\mathrm{mod}} = A_a^{\mathrm{mod}} + \rho(A_a^{\mathrm{mod}})' = \frac{\rho(1-\beta_a)}{2\sqrt{D_a}}, && 4A_a^{\mathrm{mod}}B_a^{\mathrm{mod}} = 1 - \beta_a. \end{aligned} \tag{2.12}$$

The expression for B_a^{mod} follows from $D_a' = 2\rho(1-\beta_a)$. Define the potential

$$\Phi_a^{\mathrm{mod}}(\rho) = \frac{r_4^2}{2} + \int_{r_4^2}^\rho A_a^{\mathrm{mod}}(s)\,ds,$$

and let $\omega_a^{\mathrm{mod}} = \sqrt{-1}\partial\bar\partial\Phi_a^{\mathrm{mod}}$. Its Hessian eigenvalues are A_a^{mod} and B_a^{mod}. For $a^4 < 4I_\psi$, the inequality $0 \le \beta_a \le b_a < 1$ therefore makes the form positive.

On $r \le r_1$, one has $\beta_a = m_a = 0$, so $D_a = \rho^2 + a^4$. On $r \ge r_4$, one has $m_a = a^4/2$, so $D_a = \rho^2$ and $A_a^{\mathrm{mod}} = B_a^{\mathrm{mod}} = 1/2$. Consequently,

$$\omega_a^{\mathrm{mod}} = \omega_{\mathrm{EH},a} \quad (r \le r_1), \qquad \omega_a^{\mathrm{mod}} = \omega_0 \quad (r \ge r_4). \tag{2.13}$$

The form extends smoothly across the exceptional curve, and the chosen additive constant gives $\Phi_a^{\mathrm{mod}} = \rho/2$ on the outer region.

2.5 The global explicit Kummer background. For $a^4 < 4I_\psi$, use the radial construction in the sixteen disjoint neighborhoods and the flat form elsewhere. Exact matching on $r \ge r_4$ gives a smooth global Kähler form ω_a^{mod}. Its exceptional areas are $\int_{E_\nu}\omega_a^{\mathrm{mod}} = \pi a^2$ by (2.8). Since the difference from the flat class is supported near the exceptional curves, it is a linear combination of their classes. Pairing with the curves and using (2.3) gives

$$[\omega_a^{\mathrm{mod}}] = \kappa - \frac{\pi a^2}{2}\sum_{\nu=1}^{16}[E_\nu] = \alpha_a. \tag{2.14}$$

Thus the background proves that α_a is Kähler throughout this construction range.

Extend β_a by zero outside the sixteen transition annuli. The Eguchi–Hanson volume identity and the radial coefficients give

$$\frac{(\omega_a^{\mathrm{mod}})^2}{2} = (1 - \beta_a)\mu. \tag{2.15}$$

On each annulus the quotient radial measure is $\pi^2 r^3\,dr$, so $\int_{\text{one annulus}} \beta_a\mu = \pi^2 b_a I_\psi = \pi^2 a^4/4$. Summing the sixteen contributions,

$$\int_X \beta_a\mu = 4\pi^2 a^4, \qquad \int_X \frac{(\omega_a^{\mathrm{mod}})^2}{2} = \frac{1}{2} - 4\pi^2 a^4 = \frac{\alpha_a^2}{2},$$

in agreement with (2.5).

2.6 Scale-adapted Hölder norms. Let g_a^{mod} be the metric of ω_a^{mod}. In each exceptional neighborhood, let r_ν be the orbifold radius, extended by zero on E_ν. Choose $s_a = (a^4 + r_\nu^4)^{1/4}$ near the exceptional curve and interpolate to $s_a = 1$ in a fixed outer collar, using the same fixed cutoff in all sixteen neighborhoods. The function is smooth on the resolution, since $r_\nu^4 = |t|^2(1+|\zeta|^2)^2$ in a resolution chart. It satisfies $c(a + r_\nu) \le s_a \le C(a + r_\nu)$, with constants independent of small a. Lemma 4.1 also gives

$$\mathrm{inj}_{g_a^{\mathrm{mod}}}(x) \ge c_{\mathrm{inj}} s_a(x). \tag{2.16}$$

Fix one $0 < c_* < c_{\mathrm{inj}}/2$, independent of a, small enough that s_a is uniformly comparable to $s_a(x)$ on $B_{c_* s_a(x)}(x)$.

For an integer $k \ge 0$ and $0 < \alpha < 1$, write $d_a = d_{g_a^{\mathrm{mod}}}$ and let ∇ denote the induced tensor connection. Set

$$\|T\|_{C^k_{\mathrm{sc}}} = \sum_{j=0}^{k} \sup_X s_a^j |\nabla^j T|_{g_a^{\mathrm{mod}}}, \qquad \|T\|_{C^{k,\alpha}_{\mathrm{sc}}} = \|T\|_{C^k_{\mathrm{sc}}} + [T]_{C^{k,\alpha}_{\mathrm{sc}}}.$$

With $s_a(x,y) = \min\{s_a(x), s_a(y)\}$, the seminorm is

$$[T]_{C^{k,\alpha}_{\mathrm{sc}}} = \sup_{\substack{x \ne y \\ d_a(x,y) \le c_* s_a(x,y)}} s_a(x,y)^{k+\alpha} \frac{|\nabla^k T(x) - \mathsf{P}_{yx} \nabla^k T(y)|_{g_a^{\mathrm{mod}}(x)}}{d_a(x,y)^\alpha}.$$

Here P_{yx} is parallel transport along the unique minimizing geodesic. For Hermitian endomorphisms we use the Frobenius norm. On fixed physical annuli these norms are uniformly equivalent to ordinary Hölder norms. On the core, the dilation $S_a(y) = ay$ gives $a^{-2} S_a^* g_a^{\mathrm{mod}} = g_{\mathrm{EH},1}$, and $s_a \circ S_a / a$ is uniformly comparable to one on bounded regions. The scaled norms of scalar functions and Hermitian endomorphisms are therefore uniformly equivalent to ordinary norms in these coordinates. For a tensor with p contravariant and q covariant indices, the comparison includes a^{p-q}.

2.7 The normalized Calabi–Yau residual. Let $dV_a = (\omega_a^{\mathrm{mod}})^2/2$. The target volume form is $c_a\mu$, so the normalized residual is

$$f_a := \frac{c_a\mu}{dV_a} - 1 = \frac{c_a}{1-\beta_a} - 1 = \frac{\beta_a - 8\pi^2 a^4}{1-\beta_a}. \tag{2.17}$$

The difference $f_a + 8\pi^2 a^4$ is supported in the sixteen fixed transition annuli.

Proposition 2.1 (Residual of the explicit Kummer background)**.** *For every fixed integer $k \ge 0$ and $0 < \alpha < 1$, there are constants $C_{k,\alpha}, a_{k,\alpha} > 0$ such that, for $0 < a < a_{k,\alpha}$,*

$$\|f_a\|_{C^{k,\alpha}_{\mathrm{sc}}} \le C_{k,\alpha} a^4, \qquad \int_X f_a\,dV_a = 0. \tag{2.18}$$

Proof. Since $b_a = a^4/(4I_\psi)$ and the cutoff is fixed, $\|\beta_a\|_{C^{k,\alpha}_{\mathrm{sc}}} \le C_{k,\alpha} a^4$. Indeed, its support lies in fixed physical annuli, where the scaled and ordinary norms are uniformly equivalent. For small a, one has $\|\beta_a\|_{C^0} < 1/2$; all derivatives of $s \mapsto (1-s)^{-1}$ are then uniformly bounded on its range. Formula (2.17) yields

$$\|f_a\|_{C^{k,\alpha}_{\mathrm{sc}}} \le C_{k,\alpha}\big(\|\beta_a\|_{C^{k,\alpha}_{\mathrm{sc}}} + a^4\big) \le C_{k,\alpha} a^4.$$

The mean-zero identity follows from $\int_X f_a \, dV_a = c_a \int_X \mu - \int_X dV_a = 0$, using (2.5) and (2.15). □

Remark 2.2 (The role of the parameter range). The bound $a^4 < 4I_\psi$ ensures that this background is Kähler. Convergence of the correction series is a separate conclusion of the estimates $\|G_a f_a\|_{\mathcal{X}_a} = O(a^3)$ and $\|G_a Q_a\|_{\mathrm{bil}} = O(a^{-1})$, whose product is $O(a^2)$. The resulting smallness condition is sufficient; it does not characterize the Kähler cone or the maximal convergence range of the series. In fact, on $\psi = 1$, the residual is $b_a(1 - 32\pi^2 I_\psi)/(1 - b_a)$, which can be large while the background remains positive.

2.8 The periodic Green kernel. Using real coordinates $x = (x_1, \dots, x_4)$ on $T = \mathbb{R}^4/\mathbb{Z}^4$, let $\Delta_0 = \sum_{j=1}^4 \partial_{x_j}^2$, so that $-\Delta_0$ is nonnegative. The normalized torus Green function is characterized by $-\Delta_0 G_T = \delta_0 - 1$ and $\int_T G_T \, dV_0 = 0$. Distributionally,

$$G_T(x) = \sum_{n \in \mathbb{Z}^4 \setminus \{0\}} \frac{e^{2\pi\sqrt{-1} n \cdot x}}{4\pi^2 |n|^2}.$$

For differentiation and computation we use the absolutely convergent Ewald representation of Appendix A. The normalized quotient kernel is

$$G_Y(x, y) = G_T(\widetilde{x} - \widetilde{y}) + G_T(\widetilde{x} + \widetilde{y}), \tag{2.19}$$

independent of the choices of lifts. At a two-torsion point, $G_Y(x, p_\nu) = 2G_T(\widetilde{x} - \widetilde{p}_\nu)$, with $-\Delta_0 G_Y(\,\cdot\,, p_\nu) = \delta_{p_\nu} - 2$ and $\int_Y G_Y(x, p_\nu)\, dV_0(x) = 0$. Near p_ν, it equals $1/(2\pi^2 r_\nu^2) + H_\nu(x)$, where H_ν is smooth in the orbifold covering coordinates. This kernel and its derivatives provide the exterior volume and boundary potentials in Section 3.

3 Explicit construction of the global Green operator

We now construct explicitly the normalized Green operator of the background metric introduced in Section 2. Throughout this section, $a > 0$ is fixed and sufficiently small. Write $\omega_a = \omega_a^{\mathrm{mod}}$ and $g_a = g_a^{\mathrm{mod}}$, and set $A_a = -\Lambda_{\omega_a}\sqrt{-1}\partial\bar\partial = -\frac{1}{2}\Delta_{g_a}$. The interface jump uses the ordinary outward normal derivative, so the divergence identities carry the corresponding factor two. On the flat region $A_a = A_0 = -\frac{1}{2}\Delta_0$. Our aim is to construct $G_a : \mathcal{Y}_{a,0} \to \mathcal{X}_a$ such that, for every source with $\int_X f \, dV_a = 0$,

$$A_a G_a f = f, \qquad \int_X G_a f \, dV_a = 0.$$

The construction is by domain decomposition. The exterior problem is solved using the periodic Green kernel of Section 2.8; the cap problems are solved by the separated radial kernels of Appendix B. The pieces are then joined by an explicit Dirichlet-to-Neumann matching operator.

3.1 Domain decomposition and the exterior particular solution. Fix the radius R from (2.10). Since $R > r_4$, the metric ω_a is exactly flat in a fixed neighborhood of each hypersurface $\Sigma_\nu = \{r_\nu = R\}/\{\pm 1\} \simeq S^3/\{\pm 1\}$. Write $\Sigma = \bigsqcup_{\nu=1}^{16} \Sigma_\nu$, $X_{\mathrm{out}} = Y \setminus \bigcup_{\nu=1}^{16} B_R(p_\nu)$, and let $X_{\nu,\mathrm{cap}}$ be the resolved region bounded by Σ_ν. Then

$$X = X_{\mathrm{out}} \cup_\Sigma \bigcup_{\nu=1}^{16} X_{\nu,\mathrm{cap}}. \tag{3.1}$$

For a piecewise smooth function $u = (u_{\mathrm{out}}, u_1, \dots, u_{16})$ whose traces agree on Σ, define its conormal jump by

$$(\mathcal{J}_a u)_\nu = \partial_{\nu_{\mathrm{out}}} u_{\mathrm{out}} + \partial_{\nu_{\mathrm{cap}}} u_\nu.$$

Here both normal vectors point outward from their respective domains. Because the metric is flat in a neighborhood of the interface, these are the ordinary Euclidean normal derivatives there.

A continuous piecewise A_a-harmonic function is a global weak solution precisely when $\mathcal{J}_a u = 0$.

Let G_Y be the normalized orbifold Green kernel of (2.19). Its defining equation gives

$$A_0 G_Y(\,\cdot\,, y) = \frac{1}{2}\delta_y - 1. \tag{3.2}$$

Fix one of the deleted orbifold points, say p_1, and define

$$K_{\mathrm{out}}(x,y) = 2G_Y(x,y) - 2G_Y(x,p_1), \qquad (\mathcal{P}_{\mathrm{out}} f)(x) = \int_{X_{\mathrm{out}}} K_{\mathrm{out}}(x,y) f(y)\, dV_0(y).$$

The constant terms in (3.2) cancel, and the remaining mass at p_1 lies outside X_{out}. Hence $A_0^x K_{\mathrm{out}}(x,y) = \delta_y$ and $A_0 \mathcal{P}_{\mathrm{out}} f = f$ on X_{out}. The function $\mathcal{P}_{\mathrm{out}} f$ does not in general vanish on the boundary. We next construct an explicit harmonic extension operator to remove its trace.

3.2 Periodic multipoles and exterior harmonic functions. Choose a real orthonormal basis $Y_{\ell,\gamma}$ of spherical harmonics on $S^3/\{\pm 1\}$, with even degrees $\ell = 0, 2, 4, \dots$. For $\ell \geq 2$, let $G_{\nu,\ell,\gamma}$ be the normalized periodic multipole kernel of Appendix A, so that near p_ν,

$$G_{\nu,\ell,\gamma} = r_\nu^{-\ell-2} Y_{\ell,\gamma} + O(1).$$

Since derivatives kill the constant term in the Green equation, $A_0 G_{\nu,\ell,\gamma} = 0$ on X_{out}.

For the monopole sector, the individual functions $G_Y(\cdot, p_\nu)$ are not harmonic on X_{out}, because of the constant term in (3.2). Their differences are harmonic. Define the coefficient space

$$\mathfrak{C} = \mathbb{R} \oplus \left\{ (c_{\nu,0})_{\nu=1}^{16} : \sum_{\nu=1}^{16} c_{\nu,0} = 0 \right\} \oplus \prod_{\substack{1 \leq \nu \leq 16 \\ \ell \geq 2,\ \ell \text{ even} \\ \gamma}} \mathbb{R}.$$

The first factor, with coefficient c_{const}, represents the global constant harmonic function. For $k \geq 0$, let $\mathfrak{C}^{k,\alpha}$ consist of the sequences extracted from $C^{k,\alpha}(\Sigma)$. Their boundary expansion is

$$(\mathcal{I}c)|_{\Sigma_\nu} = c_{\mathrm{const}} + c_{\nu,0} + \sum_{\ell \geq 2,\, \ell \text{ even}, \gamma} c_{\nu,\ell,\gamma} Y_{\ell,\gamma},$$

where the expansion is understood in L^2, or by Poisson regularization. The map $\mathcal{I}$ recovers the boundary function, and the coefficient norm is $\|\mathcal{I}c\|_{C^{k,\alpha}}$. The constant coefficients are unique because their sum is zero.

For $c = (c_{\mathrm{const}}, c_{\nu,0}, c_{\nu,\ell,\gamma}) \in \mathfrak{C}^{k,\alpha}$, set

$$\begin{aligned}\mathcal{M}_{\mathrm{out}} c = c_{\mathrm{const}} &+ \sum_{\nu=1}^{16} c_{\nu,0}\, 2\pi^2 R^2 G_Y(\,\cdot\,, p_\nu) \\ &+ \sum_{\substack{1\le\nu\le 16,\ \ell\ge 2 \\ \ell \text{ even},\gamma}} c_{\nu,\ell,\gamma} R^{\ell+2} G_{\nu,\ell,\gamma}.\end{aligned} \tag{3.3}$$

The relation $\sum_\nu c_{\nu,0} = 0$ makes every function in its image A_0-harmonic on X_{out}.

The factors in (3.3) are chosen so that the singular self-interaction on Σ_ν has leading trace $Y_{\ell,\gamma}$. For the monopole, $2\pi^2 R^2 G_Y = 1 + O(1)$ on its own boundary component.

Let γ_Σ be the boundary trace operator and set $\mathcal{T}_{\mathrm{out}} = \gamma_\Sigma \mathcal{M}_{\mathrm{out}}$. In the spherical basis, $\mathcal{T}_{\mathrm{out}} \mathcal{I}^{-1} = I + \mathcal{K}_{\mathrm{out}}$, where $\mathcal{K}_{\mathrm{out}}$ has a smooth kernel in the boundary variables, including the finite-rank monopole correction. Indeed, after removal of the prescribed self-singularity, Taylor estimates for the regular Green kernel and the normalization $2^\ell \ell!$ give geometric decay of the multipole columns. The restriction $R < 1/8$, while distinct centers are at least $1/2$ apart, leaves a fixed annulus of analyticity for all cross interactions. These estimates also justify convergence of the regular part of the multipole synthesis in every boundary derivative. The singular self-series is the exterior Poisson extension of its boundary function: for $r > R$, its degree-ℓ multiplier is $(R/r)^{\ell+2}$. It therefore has the prescribed $C^{k,\alpha}$ trace. This interpretation does not require sharp spherical partial sums to converge in the same Hölder norm.

Lemma 3.1 (Exterior trace isomorphism)**.** *For $k \ge 0$, the trace map $\mathcal{T}_{\mathrm{out}} : \mathfrak{C}^{k,\alpha} \to C^{k,\alpha}(\Sigma)$ is an isomorphism.*

Proof. Since $\mathcal{T}_{\mathrm{out}} \mathcal{I}^{-1} = I + \mathcal{K}_{\mathrm{out}}$ with $\mathcal{K}_{\mathrm{out}}$ smoothing, it is Fredholm of index zero.

Suppose $\mathcal{T}_{\mathrm{out}} c = 0$. Then $u = \mathcal{M}_{\mathrm{out}} c$ is harmonic on X_{out} and vanishes on its boundary. The maximum principle gives $u = 0$ on X_{out}.

Fix ν and a concentric annulus $R < r_\nu < R + \epsilon$ containing no other deleted ball. The singular self-multipoles converge there to a sum of $r_\nu^{-\ell-2} Y_{\ell,\gamma}$. The regular self-parts and the other-center contributions extend smoothly across the ball. Their sum is harmonic: the higher multipoles are harmonic, and the constant monopole Laplacians cancel because $\sum_\nu c_{\nu,0} = 0$. Uniqueness of the spherical Laurent expansion of a harmonic function on this annulus therefore forces all negative-power coefficients to vanish. This argument does not require continuation of the singular series to $r_\nu < R$. Hence $c_{\nu,\ell,\gamma} = 0, \qquad c_{\nu,0} = 0$. The remaining constant coefficient also vanishes because the boundary trace is zero. Thus $c = 0$. Hence $\mathcal{T}_{\mathrm{out}}$ is injective and therefore invertible. □

Lemma 3.2 (Spherical tails of a smoothing operator)**.** *Let K have a smooth kernel on the fixed boundary Σ. Let P_L project onto spherical degrees at most L, and put $Q_L = I - P_L$. For every integer $k \ge 0$ and $N > 0$,*

$$\|Q_L K Q_L\|_{C^{k,\alpha}(\Sigma) \to C^{k,\alpha}(\Sigma)} \le C_{k,\alpha,N} (1 + L)^{-N}. \tag{3.4}$$

The constants are uniform for a family with uniformly bounded kernel derivatives of every fixed order.

Proof. Integration by parts with the spherical Laplacian gives arbitrary polynomial decay in both indices of the matrix coefficients of K. The $C^{k,\alpha}$-norms of normalized spherical harmonics grow at most polynomially in their degree, and the multiplicity of degree ℓ is $(\ell + 1)^2$. The coefficients of an input function are bounded by its L^2-norm, hence by its

$C^{k,\alpha}$-norm. Summing the absolute norms of the output series over both degrees greater than L therefore gives (3.4), after taking sufficiently many kernel derivatives. This proof does not require the sharp spherical projections to have uniformly bounded Hölder operator norms. □

The inverse has an explicit low/high spherical representation. Write $\mathcal{T} = \mathcal{T}_{\mathrm{out}}\mathcal{I}^{-1} = I + \mathcal{K}_{\mathrm{out}}$, and let P_L project onto degrees at most L, with $Q_L = I - P_L$; the subscripts LL, LH, HL, HH denote the corresponding operator blocks. By Lemma 3.2, L may be chosen so that $\|Q_L\mathcal{K}_{\mathrm{out}}Q_L\|_{C^{2,\alpha}\to C^{2,\alpha}} < 1$. Then

$$\begin{aligned} \mathcal{T}_{HH}^{-1} &= \sum_{j=0}^{\infty}(-Q_L\mathcal{K}_{\mathrm{out}}Q_L)^j Q_L, \\ \mathcal{S}_{\mathrm{out},L} &= \mathcal{T}_{LL} - \mathcal{T}_{LH}\mathcal{T}_{HH}^{-1}\mathcal{T}_{HL}. \end{aligned} \tag{3.5}$$

The finite matrix $\mathcal{S}_{\mathrm{out},L}$ is invertible because $\mathcal{T}$ is invertible. Here all low modes, including the global constant, are retained. The usual block inverse, written out for the interface operator in (3.14), gives $\mathcal{T}^{-1}$, and $\mathcal{T}_{\mathrm{out}}^{-1} = \mathcal{I}^{-1}\mathcal{T}^{-1}$. The exterior harmonic extension is therefore $H_{\mathrm{out}}\phi = \mathcal{M}_{\mathrm{out}}\mathcal{T}_{\mathrm{out}}^{-1}\phi$. It satisfies $A_0H_{\mathrm{out}}\phi = 0$ and $\gamma_\Sigma H_{\mathrm{out}}\phi = \phi$, so the explicit zero-boundary exterior Dirichlet inverse is

$$D_{\mathrm{out}}f = \mathcal{P}_{\mathrm{out}}f - H_{\mathrm{out}}\left(\gamma_\Sigma\mathcal{P}_{\mathrm{out}}f\right).$$

Thus $A_0D_{\mathrm{out}}f = f$ and $D_{\mathrm{out}}f|_\Sigma = 0$.

3.3 The cap Dirichlet inverses. On each cap $X_{\nu,\mathrm{cap}}$, the background metric is radial. The complete separated construction is given in Appendix B; we recall only the formulas needed here.

Let $P_{p,q,\gamma}(r\theta) = r^\ell Y_{p,q,\gamma}(\theta)$ be a bihomogeneous harmonic polynomial of bidegree (p,q), where p, q are nonnegative integers and $\ell = p + q$ is even. Writing $u = w(\rho)P_{p,q,\gamma}$, where $\rho = r^2$, reduces $A_a u = F(\rho)P_{p,q,\gamma}$ to

$$-\left(\rho^{\ell+2}A_a^{\mathrm{mod}}w'\right)' + pq\,\rho^{\ell+1}(A_a^{\mathrm{mod}})'w = \rho^{\ell+1}A_a^{\mathrm{mod}}B_a^{\mathrm{mod}}F. \tag{3.6}$$

Let $w_{p,q,a}^{\mathrm{reg}}$ be the solution for which $w_{p,q,a}^{\mathrm{reg}}P_{p,q,\gamma}$ extends smoothly across the exceptional curve. Define the zero-boundary second solution by reduction of order:

$$w_{p,q,a}^{D}(\rho) = w_{p,q,a}^{\mathrm{reg}}(\rho)\int_\rho^{R^2}\frac{ds}{s^{\ell+2}A_a^{\mathrm{mod}}(s)\left(w_{p,q,a}^{\mathrm{reg}}(s)\right)^2}.$$

The radial Dirichlet kernel is $g_{p,q,a}^{D}(\rho,\sigma) = w_{p,q,a}^{\mathrm{reg}}(\rho_<)w_{p,q,a}^{D}(\rho_>)$. Expanding the source in the orthonormal angular basis gives

$$\begin{aligned} D_{\nu,a}f &= \sum_{p,q,\gamma} w_{p,q,\gamma}(\rho)P_{p,q,\gamma}(z), \\ w_{p,q,\gamma}(\rho) &= \int_0^{R^2} g_{p,q,a}^{D}(\rho,\sigma)\sigma^{\ell+1}A_a^{\mathrm{mod}}(\sigma)B_a^{\mathrm{mod}}(\sigma)F_{p,q,\gamma}(\sigma)\,d\sigma. \end{aligned}$$

Here $F_{p,q,\gamma}$ denotes the corresponding radial coefficient of the source in the polynomial normalization of Appendix B.

The harmonic extension of a boundary mode and its Dirichlet-to-Neumann eigenvalue are

$$H_{\nu,a}Y_{p,q,\gamma} = \frac{r^\ell w_{p,q,a}^{\mathrm{reg}}(r^2)}{R^\ell w_{p,q,a}^{\mathrm{reg}}(R^2)}Y_{p,q,\gamma}, \qquad \lambda_{p,q}^{\mathrm{cap}}(a) = \frac{\ell}{R} + 2R\frac{(w_{p,q,a}^{\mathrm{reg}})'(R^2)}{w_{p,q,a}^{\mathrm{reg}}(R^2)}.$$

Thus the cap inverses and their boundary operators are completely determined by explicit one-dimensional radial data.

3.4 The interface operator and its inverse. Combine the Dirichlet inverses and harmonic extensions into

$$\mathcal{D}_a = D_{\text{out}} \oplus D_{1,a} \oplus \cdots \oplus D_{16,a}, \qquad \mathcal{H}_a\phi = (H_{\text{out}}\phi, H_{1,a}\phi_1, \ldots, H_{16,a}\phi_{16}),$$

and define the global interface operator by $\mathcal{N}_a = \mathcal{J}_a\mathcal{H}_a$. For every boundary datum, Green's identity gives

$$\begin{aligned}\langle \mathcal{N}_a\phi, \phi\rangle_\Sigma = &\int_{X_{\text{out}}} |\nabla H_{\text{out}}\phi|^2_{g_0}\, dV_0 \\ &+ \sum_{\nu=1}^{16} \int_{X_{\nu,\text{cap}}} |\nabla H_{\nu,a}\phi_\nu|^2_{g_a}\, dV_a.\end{aligned} \tag{3.7}$$

The bilinear Green identity shows that $\mathcal{N}_a$ is self-adjoint with respect to the physical boundary measure. Consequently

$$\ker \mathcal{N}_a = \operatorname{span}\{\mathbf{1}\}, \tag{3.8}$$

where $\mathbf{1} = (1, \ldots, 1)$ is the common constant boundary value.

Let $\mathcal{Z}_0$ and $\mathcal{W}_0$ be the zero-boundary-mean subspaces of $C^{2,\alpha}(\Sigma)$ and $C^{1,\alpha}(\Sigma)$, respectively. The inverse below maps $\mathcal{W}_0$ to $\mathcal{Z}_0$. The splitting (3.9) makes $\mathcal{N}_a$ Fredholm of index zero, so self-adjointness and (3.8) identify its range with $\mathcal{W}_0$. Any fixed normalization of the solution by constants would be equivalent. Denote the inverse on this complement by $\mathcal{N}_a^\dagger$. To construct it explicitly, first consider a Euclidean sphere of radius R. The regular interior extension of $Y_{\ell,\gamma}$ is $(r/R)^\ell Y_{\ell,\gamma}$, with outward derivative $(\ell/R)Y_{\ell,\gamma}$; the singular exterior extension is $(R/r)^{\ell+2}Y_{\ell,\gamma}$, with outward derivative $((\ell+2)/R)Y_{\ell,\gamma}$ because the exterior normal is $-\partial_r$. Their sum has eigenvalue $2(\ell+1)/R$. Thus, defining $\mathcal{A}_\Sigma Y_{\ell,\gamma} = ((\ell+1)/R)Y_{\ell,\gamma}$ on each boundary component, we obtain

$$\mathcal{N}_a = 2\mathcal{A}_\Sigma + \mathcal{K}_a. \tag{3.9}$$

The remainder $\mathcal{K}_a$ comprises the regular periodic self-interactions, interactions between distinct boundary components, and the reflected cap terms. The periodic terms have smooth boundary kernels, and the cap terms are smoothing because the metric is exactly flat on the fixed collar $r_4 < r < R$. Their uniform estimates are proved in Section 4.

For an even integer $L \geq 0$, let P_L be the orthogonal projection onto all boundary spherical harmonics with degree $0 \leq \ell \leq L$, and put $Q_L = I - P_L$. Choose once and for all L sufficiently large that

$$\left\|(2\mathcal{A}_\Sigma)^{-1} Q_L \mathcal{K}_a Q_L\right\| < q < 1 \tag{3.10}$$

for every sufficiently small a. The existence of such an L is proved in Section 4.5.

Set $\mathcal{N}_{HH} = Q_L\mathcal{N}_a Q_L$. Its inverse is the absolutely convergent series

$$\mathcal{N}_{HH}^{-1} = \sum_{m=0}^{\infty} \left[-(2\mathcal{A}_\Sigma)^{-1} Q_L \mathcal{K}_a Q_L\right]^m (2\mathcal{A}_\Sigma)^{-1} Q_L. \tag{3.11}$$

Thus all high boundary modes are inverted by an explicitly prescribed Neumann series.

Write $\mathcal{N}_{LL}, \mathcal{N}_{LH}, \mathcal{N}_{HL}, \mathcal{N}_{HH}$ for the blocks relative to $P_L \oplus Q_L$. The finite-dimensional Schur complement is

$$\mathcal{S}_{a,L} = \mathcal{N}_{LL} - \mathcal{N}_{LH}\mathcal{N}_{HH}^{-1}\mathcal{N}_{HL}.$$

Restrict the low-mode space to $P_L\mathcal{Z}_0$, removing the common constant direction, and choose a fixed spherical basis $\{Y_I\}_{I\in\mathcal{I}_L}$ of the remaining low-mode space, orthonormal for the physical $L^2(\Sigma, d\sigma)$ inner product. Define

$$M_{a,L}[I,J] = \langle Y_I, \mathcal{S}_{a,L} Y_J\rangle_\Sigma. \tag{3.12}$$

Every matrix entry is prescribed by the periodic Ewald kernels, radial cap solutions, and high-mode series (3.11).

By (3.7), the full interface operator has no kernel apart from the global constant. Hence $M_{a,L}$ is nonsingular. Its inverse is therefore the finite algebraic expression

$$M_{a,L}^{-1} = \frac{\operatorname{adj} M_{a,L}}{\det M_{a,L}}. \tag{3.13}$$

The exterior trace inverse is treated by the same finite low-mode and convergent high-mode procedure. Thus the construction involves only finitely many finite-dimensional inversions, together with the stated convergent series.

On the complement of the common constant, self-adjointness and $\mathcal{N}_a\mathbf{1} = 0$ ensure that the blocks act between the corresponding low- and high-mode parts of the mean-zero spaces. The interface inverse is given by the block formula:

$$\mathcal{N}_a^{\dagger} = \begin{pmatrix} \mathcal{S}_{a,L}^{-1} & -\mathcal{S}_{a,L}^{-1}\mathcal{N}_{LH}\mathcal{N}_{HH}^{-1} \\ -\mathcal{N}_{HH}^{-1}\mathcal{N}_{HL}\mathcal{S}_{a,L}^{-1} & \mathcal{N}_{HH}^{-1} + \mathcal{N}_{HH}^{-1}\mathcal{N}_{HL}\mathcal{S}_{a,L}^{-1}\mathcal{N}_{LH}\mathcal{N}_{HH}^{-1} \end{pmatrix}. \tag{3.14}$$

Here $\mathcal{S}_{a,L}^{-1}$ is represented by (3.13) in the fixed low-mode basis.

3.5 The normalized global inverse. Let $f \in C^{\infty}(X)$ satisfy $\int_X f\, dV_a = 0$. Restrict f to the pieces of (3.1) and set $u^{\mathrm{D}} = \mathcal{D}_a f$. Then $A_a u^{\mathrm{D}} = f$ on each piece and $u^{\mathrm{D}}|_{\Sigma} = 0$. Seeking the solution as $u = u^{\mathrm{D}} + \mathcal{H}_a\phi$ makes the boundary values agree; the conormal matching condition $\mathcal{J}_a u = 0$ becomes

$$\mathcal{N}_a\phi = -\mathcal{J}_a\mathcal{D}_a f.$$

The right-hand side is orthogonal to the constant mode. Indeed, integrating the equations for the zero-boundary solutions and applying the divergence theorem gives

$$\langle \mathcal{J}_a\mathcal{D}_a f, \mathbf{1}\rangle_{\Sigma} = -2\int_X f\, dV_a = 0.$$

Thus $\phi = -\mathcal{N}_a^{\dagger}\mathcal{J}_a\mathcal{D}_a f$, and the global solution before normalization is

$$\widetilde{G}_a f = \mathcal{D}_a f - \mathcal{H}_a\mathcal{N}_a^{\dagger}\mathcal{J}_a\mathcal{D}_a f. \tag{3.15}$$

It satisfies $A_a\widetilde{G}_a f = f$ globally on X. Removing the additive constant gives

$$G_a f = \widetilde{G}_a f - \frac{\displaystyle\int_X \widetilde{G}_a f\, dV_a}{\displaystyle\int_X dV_a}. \tag{3.16}$$

Hence $A_a G_a f = f$ and $\int_X G_a f\, dV_a = 0$. Combining the preceding formulas gives the following explicit representation.

Theorem 3.3 (Explicit global Green representation)**.** *For every sufficiently small $a > 0$, the normalized inverse of $A_a = -\Lambda_{\omega_a^{\mathrm{mod}}}\sqrt{-1}\partial\bar{\partial}$ on the zero-mean space is given by* (3.16)*.*

More precisely, the construction uses only:

1. *the periodic Fourier–Ewald Green kernel and its derivatives;*
2. *periodic multipole kernels and spherical-harmonic projection;*
3. *one-dimensional radial Green kernels on the sixteen Kummer caps;*

4. *the high-mode Neumann series for the exterior trace and interface operators,* (3.5) *and* (3.11)*;*

5. *their finite-dimensional Schur inverses, including* (3.13)*;*

6. *finite differentiation, integration, and algebraic operations.*

For every smooth source with $\int_X f\,dV_a = 0$*, the resulting function satisfies* $A_aG_af = f$ *and* $\int_X G_af\,dV_a = 0$*.*

Remark 3.4 (No numerical certification is needed). The definition of G_a does not require a numerical lower bound for the smallest singular value of $M_{a,L}$, nor a numerically optimized high-mode contraction constant. The next section proves qualitatively that the low-mode inverse remains uniformly bounded and that one fixed sufficiently large L makes (3.10) hold uniformly for all sufficiently small a.

Remark 3.5 (The role of the appendices). Appendix A supplies the absolutely convergent Ewald representation and the normalized periodic multipoles. Appendix B supplies the radial regular solutions, Dirichlet Green kernels, harmonic extensions, and cap Dirichlet-to-Neumann coefficients. Thus no unspecified local elliptic solver enters the definition of G_a.

4 Uniform estimates for the global Green operator

We first estimate the Green operator in the scale-weighted potential norm, then obtain the bound Ca^{-1} in the unweighted complex-Hessian norm required by the Monge–Ampère equation. These norms are not uniformly equivalent as $a \to 0$. In this section, write $\omega_a = \omega_a^{\mathrm{mod}}$ and $g_a = g_a^{\mathrm{mod}}$. The interfaces lie in fixed flat collars; away from the exceptional curves the backgrounds converge smoothly to the flat metric, while on the core scale they are homothetic to the fixed Eguchi–Hanson metric.

4.1 The analytic spaces. Fix $0 < \alpha < 1$, and let $\mathcal{Y}_a = C^{0,\alpha}_{\mathrm{sc}}(X)$ with the norm of Section 2.6. Its zero-mean subspace is $\mathcal{Y}_{a,0} = \{f \in \mathcal{Y}_a : \int_X f\,dV_a = 0\}$. For a potential u, write $H_{a,u} = (\omega_a^{\mathrm{mod}})^{-1}\sqrt{-1}\partial\bar\partial u$, and equip $\mathcal{X}_a = C^{2,\alpha}_{\mathrm{sc}}(X)/\mathbb{R}$ with

$$\|u\|_{\mathcal{X}_a} = \|H_{a,u}\|_{C^{0,\alpha}_{\mathrm{sc}}}. \tag{4.1}$$

For fixed $a > 0$, this is a Banach norm equivalent to the ordinary $C^{2,\alpha}$-quotient norm: use $A_au = -\operatorname{tr} H_{a,u}$ and fixed-metric elliptic estimates on the mean-zero subspace. This equivalence may depend on a. In particular, the factor s_a^2 multiplying second derivatives in $C^{2,\alpha}_{\mathrm{sc}}$ cannot be dropped in a uniform estimate. Indeed, a nonzero compactly supported unit-core test function rescaled as $u_a(x) = a^2v(x/a)$ satisfies $\|u_a\|_{C^{2,\alpha}_{\mathrm{sc}}} = O(a^2)$, whereas $\|u_a\|_{\mathcal{X}_a} \asymp 1$ when its complex Hessian is nonzero. The test function is understood through the Eguchi–Hanson dilation and extended by zero; subtracting its average does not change this comparison.

On the fixed interface $\Sigma = \bigsqcup_{\nu=1}^{16}\Sigma_\nu$, we use the a-independent spaces $\mathcal{Z} = C^{2,\alpha}(\Sigma)$ and $\mathcal{W} = C^{1,\alpha}(\Sigma)$.

4.2 Uniform geometry in adapted charts. The following geometric input supplies the uniform local estimates.

Lemma 4.1 (Uniform adapted charts)**.** *There is a covering of* X*, with uniformly bounded overlap, such that the following hold uniformly for sufficiently small* a*.*

1. *On the outer region and the fixed transition annuli, charts have fixed physical size, and* g_a^{mod} *is uniformly equivalent in every* C^k*-norm to the Euclidean metric.*

2. *On the core* $r_\nu \le Ca$*, the change of variables* $x = ay$ *identifies* $a^{-2}g_a^{\mathrm{mod}}$ *with the unit-scale Eguchi–Hanson metric on fixed* y*-regions.*

3. *On* $Ca \le r_\nu \le r_4$*, use dyadic annuli* $\lambda \le r_\nu \le 2\lambda$*. After rescaling* $x = \lambda y$*, the metric coefficients, their inverses and all fixed-order derivatives are uniformly bounded, with uniform ellipticity constants.*

4. *The injectivity radius satisfies* (2.16)*, and* s_a *is uniformly comparable on balls of radius* $c_* s_a(x)$*.*

Proof. The first assertion follows from (2.13) and the fixed transition radii. On the core the background is exactly Eguchi–Hanson by (2.13); its homothety $x = ay$ proves the second assertion. On the intermediate region, put $\rho = r^2$. The bound $0 \le m_a(\rho) \le a^4/2$ gives $D_a(\rho) = \rho^2(1 + O(a^4/r^4))$. On a dyadic annulus $r \asymp \lambda$, rescaling by $x = \lambda y$, together with the fixed-cutoff derivative bounds for m_a, bounds every scale-invariant coefficient derivative by a universal function of $a/\lambda \le C$. Uniform ellipticity follows from

$$A_a^{\mathrm{mod}} = \frac{\sqrt{D_a}}{2\rho}, \qquad B_a^{\mathrm{mod}} = \frac{\rho(1-\beta_a)}{2\sqrt{D_a}}.$$

For the last assertion, the rescaled core is a fixed bounded region of the unit Eguchi–Hanson space, and the rescaled dyadic annuli form a smooth compact family of noncollapsed metrics including the flat limit. Together with the fixed outer charts, they have a common positive injectivity-radius lower bound; scaling back proves (2.16). Enlarge the core charts and dyadic annuli before restricting to smaller subcharts, so that the scale balls remain in regions of uniform geometric control. The explicit scale profile has bounded scale-invariant derivatives; decreasing the fixed chart fraction then gives local comparability of s_a. □

Thus the interior and boundary Schauder constants for A_a are uniform in the scale-adapted norms.

4.3 Uniform Dirichlet estimates on the caps. For the explicit cap inverse $D_{\nu,a}$ of Section 3.3, a uniform supremum estimate comes from a radial barrier. Let $\tau_a(\rho)$ be smooth across the exceptional curve, satisfy $A_a\tau_a = 1$, and vanish at $\rho = R^2$. The mode $\ell = p = q = 0$ in (3.6), followed by integration using regularity at the center, gives

$$-\big(\rho^2 A_a^{\mathrm{mod}}\tau_a'\big)' = \rho A_a^{\mathrm{mod}} B_a^{\mathrm{mod}},$$
$$-\rho^2 A_a^{\mathrm{mod}}(\rho)\tau_a'(\rho) = \int_0^\rho s A_a^{\mathrm{mod}}(s) B_a^{\mathrm{mod}}(s)\, ds.$$

By (2.12), $A_a^{\mathrm{mod}}B_a^{\mathrm{mod}} = (1-\beta_a)/4 \le 1/4$ and $A_a^{\mathrm{mod}} \ge 1/2$. Hence $0 \le -\tau_a'(\rho) \le (\int_0^\rho s/4\, ds)/(\rho^2/2) = 1/4$. Since $\tau_a(R^2) = 0$,

$$0 \le \tau_a(\rho) \le \frac{R^2}{4}. \tag{4.2}$$

Lemma 4.2 (Uniform cap Dirichlet estimate)**.** *There exists* $C > 0$*, independent of sufficiently small* a *and of* $\nu \in \{1, \dots, 16\}$*, such that*

$$\|D_{\nu,a} f\|_{C^{2,\alpha}_{\mathrm{sc}}} \le C \|f\|_{C^{0,\alpha}_{\mathrm{sc}}}. \tag{4.3}$$

Proof. The maximum principle and (4.2) yield $\|D_{\nu,a}f\|_{C^0} \le (R^2/4)\|f\|_{C^0}$. Apply the uniform local Schauder estimates of Lemma 4.1 on the core, dyadic annuli and fixed boundary collar. Bounded overlap combines these local bounds into (4.3). □

The same maximum-principle and Schauder argument for homogeneous solutions gives the corresponding extension and trace bounds.

Lemma 4.3 (Uniform cap harmonic extension)**.** *For all sufficiently small a,*

$$\|H_{\nu,a}\phi\|_{C^{2,\alpha}_{\mathrm{sc}}} \le C\|\phi\|_{C^{2,\alpha}(\Sigma_\nu)},$$
$$\|\partial_{\nu_{\mathrm{cap}}}H_{\nu,a}\phi\|_{C^{1,\alpha}(\Sigma_\nu)} \le C\|\phi\|_{C^{2,\alpha}(\Sigma_\nu)},$$

where C is independent of a.

4.4 Uniform exterior estimates. The exterior domain and its flat metric are independent of a. Fixed-domain Schauder theory therefore gives

$$\|D_{\mathrm{out}}f\|_{C^{2,\alpha}(X_{\mathrm{out}})} \le C\|f\|_{C^{0,\alpha}(X_{\mathrm{out}})},$$
$$\|H_{\mathrm{out}}\phi\|_{C^{2,\alpha}(X_{\mathrm{out}})} \le C\|\phi\|_{\mathcal{Z}},$$
$$\|\partial_{\nu_{\mathrm{out}}}H_{\mathrm{out}}\phi\|_{\mathcal{W}} \le C\|\phi\|_{\mathcal{Z}}.$$

In the following block estimates, piecewise norms mean the maximum over the closures of the seventeen pieces; normal derivatives need not match across Σ. Combining the cap and exterior bounds, and using the fixed-interface trace estimate, yields

$$\|\mathcal{D}_a f\|_{C^{2,\alpha}_{\mathrm{sc}}} \le C\|f\|_{\mathcal{Y}_a},$$
$$\|\mathcal{H}_a\phi\|_{C^{2,\alpha}_{\mathrm{sc}}} \le C\|\phi\|_{\mathcal{Z}},$$
$$\|\mathcal{J}_a u\|_{\mathcal{W}} \le C\|u\|_{C^{2,\alpha}_{\mathrm{sc}}}$$

for functions smooth up to the fixed interfaces.

4.5 Uniform smoothing of the interface remainder. Recall that $\mathcal{N}_a = 2\mathcal{A}_\Sigma + \mathcal{K}_a$ by (3.9). Its remainder has three contributions. Distinct-boundary interactions involve derivatives of the periodic Green kernel a fixed positive distance from the diagonal, so their kernels are uniformly smooth. The same-sphere kernel becomes smooth once the Euclidean singular part is removed, by Appendix A.

For the cap correction, the metric differs from the flat ball only inside $r \le r_4$. Let λ_4 be the interior Dirichlet-to-Neumann eigenvalue of a separated mode at $r = r_4$. Its energy identity gives $\lambda_4 \ge 0$. Write its radial amplitude on the flat collar as $c_+ r^\ell + c_- r^{-\ell-2}$. Matching the logarithmic derivative at r_4 yields

$$\frac{c_-}{c_+}r_4^{-2\ell-2} = \frac{\ell - r_4\lambda_4}{\ell + 2 + r_4\lambda_4}, \qquad \left|\frac{c_-}{c_+}r_4^{-2\ell-2}\right| \le 1.$$

In particular $c_+ \neq 0$. With $\vartheta_* = r_4/R < 1$, this implies

$$|\mathfrak{r}_{p,q}(a)| \le \vartheta_*^{2\ell+2}, \qquad \left|\lambda^{\mathrm{cap}}_{p,q}(a) - \frac{\ell}{R}\right| \le \frac{2(\ell+1)}{R}\frac{\vartheta_*^{2\ell+2}}{1-\vartheta_*^{2\ell+2}}.$$

Thus the cap correction is uniformly smoothing. For the exterior, the smoothing trace correction and its bounded inverse show that the Dirichlet-to-Neumann operator differs from the singular Euclidean symbol by a fixed smoothing operator. We conclude:

Lemma 4.4 (Uniform smoothing)**.** *For every $m \geq 0$, the spherical matrix coefficients satisfy*

$$|\langle Y_I, \mathcal{K}_a Y_J\rangle_\Sigma| \leq C_m(1+\ell_I+\ell_J)^{-m},$$

where C_m is independent of sufficiently small a.

This decay and Lemma 3.2, applied to $(2\mathcal{A}_\Sigma)^{-1}\mathcal{K}_a$, show that the high-mode tails tend to zero in the $\mathcal{Z} \to \mathcal{Z}$ operator norm, uniformly for $0 < a < a_0$. Choose a fixed sufficiently large even L so that

$$\sup_{0<a<a_0} \left\|(2\mathcal{A}_\Sigma)^{-1} Q_L \mathcal{K}_a Q_L\right\| \leq q < 1. \tag{4.4}$$

The series (3.11) then gives

$$\|\mathcal{N}_{HH}^{-1}\|_{\mathcal{W}\to\mathcal{Z}} \leq C \tag{4.5}$$

uniformly for sufficiently small a.

4.6 Uniform control of the low modes. The fixed exterior controls all boundary directions except the common constant, so no flat-orbifold cap limit is needed here. Let $\Lambda_{\mathrm{out}}\phi = \partial_{\nu_{\mathrm{out}}} H_{\mathrm{out}}\phi$, and use the Sobolev norms on Σ with its physical measure.

Lemma 4.5 (Coercivity of the fixed exterior)**.** *Every $\phi \in \mathcal{Z}_0$ satisfies*

$$\langle \Lambda_{\mathrm{out}}\phi, \phi\rangle_\Sigma \geq c_0 \|\phi\|^2_{H^{1/2}(\Sigma)}, \tag{4.6}$$

for some $c_0 > 0$ independent of a.

Proof. Put $v = H_{\mathrm{out}}\phi$. Its trace has zero boundary mean on the fixed connected exterior domain, so the Poincaré inequality gives $\|v\|_{H^1} \leq C\|\nabla v\|_{L^2}$. Indeed, failure would give a sequence with unit H^1-norm and gradients tending to zero. Compactness in L^2 and continuity of the boundary trace mean force the limit to be a constant of zero boundary mean, hence zero, contradicting the normalization. The fixed-domain trace theorem now gives

$$\|\phi\|_{H^{1/2}(\Sigma)} \leq C\|v\|_{H^1(X_{\mathrm{out}})} \leq C\|\nabla v\|_{L^2(X_{\mathrm{out}})}.$$

Green's identity identifies the squared gradient norm with $\langle \Lambda_{\mathrm{out}}\phi, \phi\rangle_\Sigma$, proving the claim. □

The cap energies are nonnegative. Hence (3.7) and (4.6) imply the uniform bound $\langle \mathcal{N}_a\phi, \phi\rangle_\Sigma \geq c_0\|\phi\|^2_{H^{1/2}(\Sigma)}$ for $\phi \in \mathcal{Z}_0$.

Lemma 4.6 (Uniform low-mode inverse)**.** *For the fixed cutoff L in* (4.4)*,*

$$\|M_{a,L}^{-1}\| \leq C_L \tag{4.7}$$

uniformly for sufficiently small a, after removal of the global constant direction.

Proof. For $x \in P_L\mathcal{Z}_0$, put $y = -\mathcal{N}_{HH}^{-1}\mathcal{N}_{HL}x$ and $\psi = x + y$. The inverse exists by (4.5). Since y contains only high spherical modes, it has zero boundary mean and is $L^2(\Sigma)$-orthogonal to x; thus $\psi \in \mathcal{Z}_0$. By construction, $Q_L\mathcal{N}_a\psi = 0$ and $P_L\mathcal{N}_a\psi = \mathcal{S}_{a,L}x$, so

$$\begin{aligned}\langle \mathcal{S}_{a,L}x, x\rangle_\Sigma &= \langle \mathcal{N}_a\psi, \psi\rangle_\Sigma \\ &\geq c_0\|\psi\|^2_{H^{1/2}(\Sigma)} \geq c_1\|x\|^2_{L^2(\Sigma)}.\end{aligned} \tag{4.8}$$

The last inequality uses the fixed Sobolev embedding into L^2 and the orthogonality of x and y. Thus every eigenvalue of $M_{a,L}$, represented in the fixed physical L^2-orthonormal basis, has a uniform positive lower bound. Its inverse is uniformly bounded; equivalence of norms in this fixed finite-dimensional space proves (4.7). □

The off-diagonal blocks are uniformly bounded, since the free symbol is diagonal and the remainders are uniformly smoothing. Combining (4.5), (4.7) and the block formula (3.14) gives:

Proposition 4.7 (Uniform interface inverse)**.** *For sufficiently small a, $\|\mathcal{N}_a^{\dagger}\|_{\mathcal{W}_0\to\mathcal{Z}_0} \le C$, with C independent of a.*

4.7 The global estimates and their different norms. The block Dirichlet, harmonic-extension and fixed-interface trace estimates, together with Proposition 4.7, give $\|\widetilde{G}_a f\|_{C^{2,\alpha}_{\mathrm{sc}}} \le C\|f\|_{\mathcal{Y}_a}$ in (3.15). Matching traces and normal derivatives produces a global weak solution; local elliptic regularity across the fixed collars gives the same estimate there. Subtracting the average changes its supremum by at most a factor two, and therefore

$$\|G_a f\|_{C^0} \le C\|f\|_{\mathcal{Y}_a}. \tag{4.9}$$

These bounds also extend the inverse to $f \in \mathcal{Y}_{a,0}$. For fixed a, choose smooth mean-zero f_j such that $\|f_j\|_{\mathcal{Y}_a} \le 2\|f\|_{\mathcal{Y}_a}$ and $f_j \to f$ in $C^{0,\alpha'}$, where $0 < \alpha' < \alpha$. The preceding estimates give uniform $C^{2,\alpha}$ bounds for $G_a f_j$, with constants allowed to depend on this fixed a. Compactness and the equation give a mean-zero $C^{2,\alpha}$ limit satisfying $A_a u = f$. Uniqueness makes it independent of the approximation, and the bounds pass to the limit; smooth density in the full $C^{0,\alpha}$ norm is not needed. These estimates control weighted potential derivatives. The stronger norm $\mathcal{X}_a$ requires the following gradient bound.

Lemma 4.8 (Uniform gradient bound)**.** *If $u = G_a f$, then*

$$\|\nabla u\|_{C^0} \le C\big(\|u\|_{C^0} + \|f\|_{C^0}\big) \le C\|f\|_{\mathcal{Y}_a}. \tag{4.10}$$

Proof. The Ricci tensor has a uniform lower bound $\mathrm{Ric}(g_a) \ge -Kg_a$: it vanishes in the Eguchi–Hanson and flat regions, while on the fixed transition annuli this follows from $\mathrm{Ric}(\omega_a) = -\sqrt{-1}\partial\bar\partial \log(1-\beta_a)$ and uniform metric equivalence. For the proof only, put $P_t = e^{-tA_a}$. Bochner's identity and the maximum principle give $|\nabla P_t h|^2 \le e^{Kt} P_t|\nabla h|^2$. The variance identity then yields

$$\begin{aligned} P_t(h^2) - (P_t h)^2 &= \int_0^t P_s|\nabla P_{t-s} h|^2\, ds \\ &\ge \left(\int_0^t e^{-Ks}\, ds\right) |\nabla P_t h|^2. \end{aligned}$$

Positivity of P_t gives $\|\nabla P_t h\|_{C^0} \le Ct^{-1/2}\|h\|_{C^0}$ for $0 < t \le 1$, with C independent of a. The identity $u = P_1 u + \int_0^1 P_t f\, dt$ proves the first inequality in (4.10); (4.9) proves the second. The heat semigroup is used only for this estimate, and supplies no additional kernel to the explicit definition of G_a. □

Theorem 4.9 (Sufficient Green estimate)**.** *For sufficiently small $a > 0$ and $f \in \mathcal{Y}_{a,0}$,*

$$\|G_a f\|_{\mathcal{X}_a} \le Ca^{-1}\|f\|_{\mathcal{Y}_a}, \tag{4.11}$$

with C independent of a.

Proof. Put $u = G_a f$. Take a chart centered at x of physical radius comparable to $s = s_a(x)$, and subtract $u(x)$ before rescaling. Lemma 4.8 gives $\|u - u(x)\|_{C^0(B_{cs}(x))} \le Cs\|f\|_{\mathcal{Y}_a}$. The Schauder estimate on the rescaled chart, whose metric coefficients are uniformly bounded, yields

$$\begin{aligned} \|H_{a,u}\|_{C^{0,\alpha}_{\mathrm{sc}}(B_{c's}(x))} &\le C\left(s^{-2}\|u - u(x)\|_{C^0(B_{cs}(x))} + \|f\|_{\mathcal{Y}_a}\right) \\ &\le C(1 + s^{-1})\|f\|_{\mathcal{Y}_a}. \end{aligned}$$

Since $s_a \geq ca$, take the supremum over the adapted covering. Uniform rescaled geometry gives the same control of the parallel-transport Hölder seminorm. □

Remark 4.10 (The estimate actually used). The bound $\|G_a\|_{\mathcal{Y}_{a,0}\to C^{2,\alpha}_{\mathrm{sc}}} \leq C$ is distinct from (4.11). No uniform bound in the unweighted Hessian norm is asserted; a qualitative absence of new constant modes does not establish one. Combining (4.11) with the residual estimate gives $\|G_a f_a\|_{\mathcal{X}_a} = O(a^3)$, as required below.

5 The recursive representation of the Calabi–Yau potential

We prove that the explicitly prescribed recursion represents the normalized potential v_a in (1.2). The convergence estimates use only the background residual, the Green estimate, and the quadratic volume identity. The Calabi–Yau theorem is used to identify the limit and inherit its smoothness.

5.1 A convergent quadratic recursion. We first record the elementary Banach-space argument, which will also be used in the elliptic discussion.

Proposition 5.1 (Convergent quadratic recursion)**.** *Let $\mathcal{E}$ be a real Banach space, $F \in \mathcal{E}$, and $\mathcal{B} : \mathcal{E} \times \mathcal{E} \to \mathcal{E}$ a bounded bilinear map. Set $\eta = \|F\|_{\mathcal{E}}$ and $b = \|\mathcal{B}\|_{\mathrm{bil}}$, and suppose $4\eta b < 1$. Define*

$$V_1 = F, \qquad V_n = \sum_{j=1}^{n-1} \mathcal{B}(V_j, V_{n-j}), \qquad n \geq 2.$$

Then

$$\|V_n\|_{\mathcal{E}} \leq \mathrm{Cat}_{n-1}\, \eta^n b^{n-1}, \tag{5.1}$$

with the right-hand side interpreted as η for $n = 1$. The series $u = \sum_{n\geq 1} V_n$ converges absolutely, satisfies $u = F + \mathcal{B}(u, u)$, and obeys

$$\|u\|_{\mathcal{E}} \leq \frac{2\eta}{1 + \sqrt{1 - 4\eta b}} \leq 2\eta. \tag{5.2}$$

Proof. The Catalan identity $\mathrm{Cat}_{n-1} = \sum_{j=1}^{n-1} \mathrm{Cat}_{j-1}\,\mathrm{Cat}_{n-j-1}$ proves (5.1) by induction. The generating function $\sum_{m\geq 0} \mathrm{Cat}_m\, t^m = 2/(1 + \sqrt{1 - 4t})$, valid for $|t| < 1/4$, gives absolute convergence and (5.2). The double series for $\mathcal{B}(u, u)$ converges absolutely, since the sum of its norms is at most $b(\sum_n \|V_n\|_{\mathcal{E}})^2$. Regrouping it by total degree gives $\mathcal{B}(u, u) = \sum_{n\geq 2} V_n = u - F$. □

5.2 The quadratic volume identity. For the remainder of the section write $\omega_a = \omega_a^{\mathrm{mod}}$, $dV_a = \omega_a^2/2$, and $A_a = -\Lambda_{\omega_a}\sqrt{-1}\partial\bar\partial$. Recall that $\mathcal{X}_a$ is the potential space modulo constants with norm (4.1), and $\mathcal{Y}_{a,0}$ is the zero-mean source space. The quadratic term is polarized by $Q_a(u, v) = \frac{\sqrt{-1}\partial\bar\partial u \wedge \sqrt{-1}\partial\bar\partial v}{\omega_a^2}$. In an ω_a-unitary frame, let U and V represent the relative complex Hessians of u and v, respectively. Then

$$Q_a(u, v) = \frac{1}{2}\big((\mathrm{tr}U)(\mathrm{tr}V) - \mathrm{tr}(UV)\big) = \frac{1}{2}\langle(\mathrm{tr}U)I - U, V\rangle_F.$$

For 2×2 Hermitian matrices, $|(\mathrm{tr}U)I - U|_F = |U|_F$. This gives the pointwise estimate $|Q_a(u, v)| \leq \frac{1}{2}|U|_F|V|_F$. Taking differences after parallel transport gives the corresponding Hölder product bound.

Proposition 5.2 (Quadratic estimate and compatibility)**.** *The map $Q_a : \mathcal{X}_a \times \mathcal{X}_a \to \mathcal{Y}_{a,0}$ is symmetric and bilinear, with*

$$\|Q_a(u,v)\|_{\mathcal{Y}_a} \le \frac{1}{2}\|u\|_{\mathcal{X}_a}\|v\|_{\mathcal{X}_a}. \tag{5.3}$$

In particular, every quadratic source is compatible with G_a.

Proof. The product estimate just proved controls the supremum and the scale-adapted Hölder seminorm. Moreover,

$$\int_X Q_a(u,v)\,dV_a = \frac{1}{2}\int_X \sqrt{-1}\partial\bar\partial u \wedge \sqrt{-1}\partial\bar\partial v = 0.$$

For smooth potentials this is Stokes' theorem. Approximation in C^2 gives the identity for $C^{2,\alpha}$ potentials. □

The volume expansion and the prescribed volume equation give

$$\begin{aligned}\frac{(\omega_a + \sqrt{-1}\partial\bar\partial u)^2/2}{dV_a} &= 1 + \Lambda_{\omega_a}\sqrt{-1}\partial\bar\partial u + Q_a(u,u),\\ -A_a u + Q_a(u,u) = f_a, &\qquad u = -G_a f_a + G_a Q_a(u,u).\end{aligned}$$

These identities hold for mean-zero $C^{2,\alpha}$ potentials; positivity will follow by continuity along the convergent series.

5.3 The explicit coefficients and their convergence. Set $F_a = -G_a f_a$, $\eta_a = \|F_a\|_{\mathcal{X}_a}$, and $\mathfrak{b}_a = \|G_a Q_a\|_{\mathrm{bil}}$. Theorem 4.9 and Proposition 2.1, together with (5.3), give

$$\eta_a \le C_1 a^3, \qquad \mathfrak{b}_a \le C_2 a^{-1}. \tag{5.4}$$

After decreasing a_0, we therefore have

$$4\eta_a\mathfrak{b}_a \le 4C_1C_2a^2 < 1. \tag{5.5}$$

For the coefficients in (1.3), the zero-mean identity ensures that every application of G_a is defined. Applying Proposition 5.1 with $\mathcal{E} = \mathcal{X}_a$ and $\mathcal{B} = G_aQ_a$ gives

$$\|U_{a,n}\|_{\mathcal{X}_a} \le \mathrm{Cat}_{n-1}\,\eta_a^n\mathfrak{b}_a^{n-1} \le \mathrm{Cat}_{n-1}\,C_1^nC_2^{n-1}a^{2n+1}. \tag{5.6}$$

Consequently,

$$u_a := \sum_{n=1}^{\infty} U_{a,n}, \qquad \|u_a\|_{\mathcal{X}_a} \le \frac{2\eta_a}{1+\sqrt{1-4\eta_a\mathfrak{b}_a}} \le 2\eta_a \le Ca^3. \tag{5.7}$$

All representatives are chosen to have zero dV_a-mean. For fixed a, the norm equivalence in Section 4.1 gives convergence in $C^{2,\alpha}(X)$, and $\int_X u_a\,dV_a = 0$.

Repeated substitution gives $U_{a,n} = \sum_{T\in\mathfrak{T}_n} U_{a,T}$, where $\mathfrak{T}_n$ consists of rooted planar full binary trees with n leaves and has cardinality Cat_{n-1}. Each leaf is assigned F_a, and each internal vertex applies G_aQ_a to its two inputs. This determines every coefficient through the explicit background Green representation.

5.4 Identification with the Calabi–Yau potential. For a fixed a satisfying (5.5), define $u_a(t) = \sum_{n\geq 1} t^n U_{a,n}$ for $0 \leq t \leq 1$. The Catalan majorant gives uniform convergence in $C^{2,\alpha}$ on this interval, so $u_a(t)$ is continuous in t and $u_a(0) = 0$. The volume identity $c_a \int_X \mu = \int_X dV_a > 0$ ensures $c_a > 0$. Absolute convergence and bounded bilinearity give

$$u_a(t) = -tG_a f_a + G_a Q_a(u_a(t), u_a(t)),$$
$$\frac{(\omega_a + \sqrt{-1}\partial\bar\partial u_a(t))^2/2}{dV_a} = 1 + tf_a = (1-t) + t\frac{c_a\mu}{dV_a} > 0.$$

The last ratio is the determinant of the Hermitian endomorphism $\omega_a^{-1}(\omega_a + \sqrt{-1}\partial\bar\partial u_a(t))$. At each point of X, its eigenvalues vary continuously, start positive at $t = 0$, and cannot pass through zero. At $t = 1$, the convergence condition gives

$$\widetilde{\omega}_a := \omega_a + \sqrt{-1}\partial\bar\partial u_a > 0, \qquad \frac{\widetilde{\omega}_a^2}{2} = c_a\mu. \tag{5.8}$$

The form has $C^{0,\alpha}$ coefficients and belongs to α_a. For fixed a, compactness gives a positive lower bound relative to ω_a.

Let v_a be the smooth normalized potential given by (1.2), and put $h = u_a - v_a$. Since the two volume forms agree,

$$\sqrt{-1}\partial\bar\partial h \wedge (\widetilde{\omega}_a + \omega_{\mathrm{CY},a}) = \widetilde{\omega}_a^2 - \omega_{\mathrm{CY},a}^2 = 0.$$

Integration by parts gives

$$0 = \int_X h\,\sqrt{-1}\partial\bar\partial h \wedge (\widetilde{\omega}_a + \omega_{\mathrm{CY},a}) = -\int_X dh \wedge d^c h \wedge (\widetilde{\omega}_a + \omega_{\mathrm{CY},a}).$$

This identity is valid at the present regularity: approximate u_a by smooth functions in C^2, apply Stokes' theorem to the resulting closed forms, and pass to the limit. With our convention, $dh \wedge d^c h = \sqrt{-1}\partial h \wedge \bar\partial h$ is nonnegative. The two forms in the energy identity are positive on the compact surface, by (5.8) and (1.2); hence $dh = 0$. Both potentials have zero dV_a-mean, so

$$u_a = v_a, \qquad \widetilde{\omega}_a = \omega_{\mathrm{CY},a}. \tag{5.9}$$

In particular, the sum is smooth and represents the unique Ricci-flat metric in α_a. No estimate for the Yau potential was needed to establish the convergence.

Completion of the proof of Theorem 1.1. The background lies in α_a by (2.14), and Theorem 3.3 specifies the normalized Green operator in the recursion. The estimates (5.4)–(5.6) prove absolute convergence and the coefficient bound. Equations (5.8)–(5.9) identify the sum with v_a. The metric representation and the bound $\|v_a\|_{\mathcal{X}_a} \leq Ca^3$ now follow from (5.7). This proves all assertions. □

Remark 5.3 (Scope of convergence). The smallness condition (5.5) is sufficient for convergence, and the estimates verify it near the Kummer limit $a = 0$. The series is summed for fixed $a > 0$; convergence throughout the full range in which α_a is Kähler is not asserted. Smoothness of the limit follows from the comparison with the Calabi–Yau solution. It does not by itself prove convergence of the same finite sums in every C^k-norm or provide higher-derivative truncation bounds. The metric tail estimate proved below is in $C^{0,\alpha}_{\mathrm{sc}}$.

6 Evaluation and truncation

The preceding sections represent the Calabi–Yau metric by a convergent explicit series. We now record truncation formulas and the additional bounds needed for certified numerical evaluation. Throughout this section, write $\omega_a = \omega_a^{\mathrm{mod}}$.

No part of this section is needed for the proof of Theorem 1.1. Its purpose is to make precise the computational content of the word "explicit."

The calculation proceeds from the explicit background and its residual to the Green operator and then to the binary-tree correction. Optional symmetry reductions, normalized angular bases, and Fourier moments are collected in Appendix C.

6.1 The background and its residual. The background is evaluated from the fixed cutoff profile and the radial formulas (2.11)–(2.12). They require one-dimensional quadrature and elementary operations. Together with the residual formula (2.17), they give

$$\beta_a = a^4\beta^{(4)}, \qquad \beta^{(4)}(\rho) = \frac{\psi(\sqrt{\rho})}{4I_\psi},$$
$$a^{-4}f_a \longrightarrow f^{(4)} := \beta^{(4)} - 8\pi^2.$$

The convergence holds in every fixed Hölder norm on the transition region. Thus both the background and its leading forcing are prescribed explicitly.

6.2 Evaluation of the periodic kernel. The periodic Green kernel is evaluated through (A.4). For each Ewald parameter $\sigma > 0$, the real-space and Fourier sums and their differentiated sums have Gaussian tails. Truncating at $|m| \le M$ and $|n| \le N$ gives

$$\left|\partial^\beta G_T - \partial^\beta G_T^{M,N}\right| \le C_\beta \left[(1+M)^{C_\beta} e^{-cM^2/\sigma} + (1+N)^{C_\beta} e^{-c\sigma N^2}\right].$$

Here the constants are for a fixed Ewald parameter and a fixed compact set away from the lattice singularities. The same-sphere singularity must first be separated analytically as in Appendix A. Evaluation of boundary matrix elements also requires quadrature and angular truncation estimates; the Ewald tail alone does not certify the complete boundary computation.

6.3 Evaluation of the cap kernels. For every pair (p, q), the cap calculation begins with the regular radial solution $w^{\mathrm{reg}}_{p,q,a}$. Near the exceptional curve it is initialized by its convergent Frobenius expansion, evaluated at $0 < \rho_* < \min\{a^2, r_1^2\}$, from Lemma B.2. It is then continued across the radial transition region by the one-dimensional equation (B.5).

The reduction-of-order formula (B.9) gives the zero-boundary second solution by one-dimensional quadrature. The product (B.10) then gives the radial Green kernel, and (B.14) gives the cap Dirichlet-to-Neumann coefficient from the boundary value and derivative of the regular solution. The continuation is specified by the convergent iterated-integral series (B.8); a numerical ODE solver is one way to evaluate it. Thus the radial solution is not left as an unspecified local elliptic inverse.

6.4 Evaluation of the interface inverse. Choose an even spherical cutoff L for which the high-mode contraction in (4.4) holds.

The finite low-mode matrix $M_{a,L}$ is assembled from Ewald boundary contractions, cap Dirichlet-to-Neumann coefficients, and high-mode Schur corrections.

Its dimension is finite and independent of the desired numerical accuracy. The adjugate formula (3.13) is its exact symbolic inverse; numerical evaluation may use a stable factorization. For the high modes, put $K_{a,L} = (2\mathcal{A}_\Sigma)^{-1} Q_L \mathcal{K}_a Q_L$. Then $\|K_{a,L}\| \le q < 1$, and (3.11)

gives a convergent Neumann series. Retaining the terms indexed by $0, \dots, J$ gives

$$\left\| \mathcal{N}_{HH}^{-1} - \sum_{j=0}^{J} (-K_{a,L})^j (2\mathcal{A}_\Sigma)^{-1} Q_L \right\| \leq \frac{q^{J+1}}{1-q} \, \|(2\mathcal{A}_\Sigma)^{-1} Q_L\|.$$

Hence the interface matching may be evaluated with a geometric tail bound.

6.5 The first nonlinear correction. The first nonlinear term is $U_{a,2} = G_a Q_a(U_{a,1}, U_{a,1})$, where $U_{a,1} = -G_a f_a$. The estimates give $\|U_{a,1}\|_{\mathcal{X}_a} = O(a^3)$ and $\|U_{a,2}\|_{\mathcal{X}_a} = O(a^5)$. The general coefficient bound is (5.6). These upper bounds suffice for convergence and need not be sharp; the estimate used here does not imply $U_{a,n} = O(a^{4n})$ globally.

6.6 Truncation of the binary-tree series. Put $t_a = \eta_a \mathfrak{b}_a$ and $u_a^{[M]} = \sum_{n=1}^{M} U_{a,n}$. For an admissible a, $4t_a < 1$. Using $\mathrm{Cat}_m \leq 4^m$ and summing the geometric majorant gives

$$\|u_a - u_a^{[M]}\|_{\mathcal{X}_a} \leq \eta_a \frac{(4t_a)^M}{1-4t_a}. \tag{6.1}$$

Once upper bounds for η_a and $\mathfrak{b}_a$ are available, this gives a stopping criterion for the exact nonlinear series.

6.7 Scope of the numerical error estimates. For fixed admissible a, the Catalan bound controls the exact tree truncation in $\mathcal{X}_a$, hence the metric error in $C^{0,\alpha}_{\mathrm{sc}}$. The Ewald and Volterra formulas and the high-mode Neumann series give separate convergent representations of their respective components. To turn these formulas into a certified computation at a specified numerical value of a, one must bound the constants and propagate the errors of the components. In particular, a finite Neumann truncation still acts on infinitely many angular modes and must be accompanied by an angular cutoff.

Proposition 6.1 (Conditional error certificate)**.** *Suppose η_a and $\mathfrak{b}_a$ have certified upper bounds $\overline{\eta}_a$ and $\overline{b}_a$, with $4\overline{\eta}_a\overline{b}_a < 1$. If computed smooth, mean-zero terms $\widetilde{U}_{a,n}$, $1 \leq n \leq M$, satisfy $\|\widetilde{U}_{a,n} - U_{a,n}\|_{\mathcal{X}_a} \leq \epsilon_n$, then*

$$\left\| \omega_a^{-1} \left(\omega_a + \sqrt{-1}\partial\bar{\partial} \sum_{n=1}^{M} \widetilde{U}_{a,n} - \omega_{\mathrm{CY},a} \right) \right\|_{C^{0,\alpha}_{\mathrm{sc}}} \leq \sum_{n=1}^{M} \epsilon_n + \overline{\eta}_a \frac{(4\overline{\eta}_a\overline{b}_a)^M}{1-4\overline{\eta}_a\overline{b}_a}.$$

Proof. Apply the triangle inequality and (6.1). The positive scalar majorant is increasing in both of its arguments. □

Certification of the ϵ_n requires quadrature and angular tail bounds, radial continuation errors, bounds for the inverse matrices, and error propagation through the finite tree recursion. For example, if a matrix approximation satisfies $\|\widetilde{M}^{-1}\| \leq K$ and $\|M - \widetilde{M}\| \leq \epsilon < K^{-1}$, then $\|M^{-1} - \widetilde{M}^{-1}\| \leq K^2\epsilon/(1-K\epsilon)$. The preceding proofs establish finite analytic constants and a positive convergence threshold. Numerical upper bounds for these constants, a certified numerical value of a_0, and certified component errors remain to be determined; an interval-arithmetic implementation is not supplied here. Likewise, smoothness of the exact solution does not by itself give a C^k error bound for arbitrary finite approximations: such a bound requires a differentiated convergence argument or sufficiently strong bounds for their residuals. The proved truncation certificate above is in the displayed $C^{0,\alpha}_{\mathrm{sc}}$ metric norm.

7 The elliptic K3 setting

Proposition 5.1 also clarifies how the nonlinear construction can be applied to collapsing elliptic K3 surfaces. Gross–Wilson [5] construct approximate Ricci-flat metrics by gluing two complementary local models. On the smooth locus, the *semi-flat metric* [4] is an explicit Ricci-flat Kähler metric with flat, translation-invariant elliptic fibers; the periods determine its variation over the base [5, Example 2.2]. It models the collapsing regular fibers but does not extend smoothly across the nodal fibers.

Near each I_1 fiber, the *Ooguri–Vafa (OV) metric* [9] provides an explicit hyper-Kähler model, built from a periodic Gibbons–Hawking potential, that is smooth on the total space across the nodal fiber [5, Section 3]. The nonzero Fourier modes of this potential decay exponentially on fixed matching annuli as the fiber area tends to zero, while its zero mode produces the corresponding semi-flat model [5, Lemma 3.1]. Matching the periods and translation data and gluing these OV models to the semi-flat metric yields a smooth global Kähler background with exponentially small Monge–Ampère residual [5, Theorem 4.5]. This exponential accuracy, together with the Green-operator bounds assumed below, is the relevant feature for the recursion.

For this discussion, let $\pi : X \to \mathbb{P}^1$ be an elliptic K3 surface with zero section S and twenty-four singular fibers of type I_1. Write $[F]$ for the fiber class and consider $\alpha_\varepsilon = \varepsilon[S]+(\varepsilon^{-1}+\varepsilon)[F]$, so that $\alpha_\varepsilon^2/2 = 1$. Normalize $\mu = \frac{1}{4}\Omega \wedge \bar{\Omega}$ by $\int_X \mu = 1$. For a smooth Kähler background $\widehat{\omega}_\varepsilon \in \alpha_\varepsilon$, put $dV_\varepsilon = \widehat{\omega}_\varepsilon^2/2$ and define

$$A_\varepsilon = -\Lambda_{\widehat{\omega}_\varepsilon}\sqrt{-1}\partial\bar{\partial}, \qquad f_\varepsilon = \frac{\mu}{dV_\varepsilon} - 1,$$
$$Q_\varepsilon(u,v) = \frac{\sqrt{-1}\partial\bar{\partial}u \wedge \sqrt{-1}\partial\bar{\partial}v}{\widehat{\omega}_\varepsilon^2}.$$

Both f_ε and $Q_\varepsilon(u,v)$ have zero dV_ε-mean. Let $\mathbf{G}_\varepsilon$ denote the inverse of A_ε on functions of zero dV_ε-mean, with its output normalized to have zero mean as well. The exact equation is then $u = -\mathbf{G}_\varepsilon f_\varepsilon + \mathbf{G}_\varepsilon Q_\varepsilon(u,u)$.

Fix a smooth reference metric g_*, and let $\mathcal{E}_\varepsilon$ be the real $H^5(X, g_*)$ functions of zero dV_ε-mean, with the $H^5(X, g_*)$ norm. Suppose that, for constants $C, c > 0$ and $M \geq 0$, the following bounds hold for every sufficiently small $\varepsilon > 0$:

$$\|\mathbf{G}_\varepsilon f_\varepsilon\|_{\mathcal{E}_\varepsilon} \leq Ce^{-c/\varepsilon}, \qquad \|\mathbf{G}_\varepsilon Q_\varepsilon\|_{\mathrm{bil}} \leq C\varepsilon^{-M}, \tag{7.1}$$
$$\|\widehat{\omega}_\varepsilon^{-1}\sqrt{-1}\partial\bar{\partial}v\|_{C^0} \leq C\varepsilon^{-M}\|v\|_{\mathcal{E}_\varepsilon} \qquad \text{for } v \in \mathcal{E}_\varepsilon. \tag{7.2}$$

The bilinear norm in (7.1) is on $\mathcal{E}_\varepsilon \times \mathcal{E}_\varepsilon \to \mathcal{E}_\varepsilon$. The endomorphism norm in (7.2) is measured with respect to $\widehat{\omega}_\varepsilon$. An exponentially small residual in $H^3(X, g_*)$, together with a polynomial bound for $\mathbf{G}_\varepsilon : H^3_0 \to H^5_0$, yields the first estimate, after decreasing the exponential rate; the subscript denotes zero dV_ε-mean.

These assumptions give $4\|\mathbf{G}_\varepsilon f_\varepsilon\|_{\mathcal{E}_\varepsilon}\|\mathbf{G}_\varepsilon Q_\varepsilon\|_{\mathrm{bil}} \to 0$. Consequently, the recursion

$$U_{\varepsilon,1} = -\mathbf{G}_\varepsilon f_\varepsilon, \qquad U_{\varepsilon,n} = \mathbf{G}_\varepsilon \sum_{j=1}^{n-1} Q_\varepsilon(U_{\varepsilon,j}, U_{\varepsilon,n-j}), \quad n \geq 2,$$

converges in $\mathcal{E}_\varepsilon$, with sum u_ε of norm $O(e^{-c/\varepsilon})$, and solves $(\widehat{\omega}_\varepsilon + \sqrt{-1}\partial\bar{\partial}u_\varepsilon)^2/2 = \mu$. The homotopy $\sum_{n\geq 1} t^n U_{\varepsilon,n}$ has volume ratio $(1-t) + t\mu/dV_\varepsilon > 0$ for $0 \leq t \leq 1$; it therefore preserves positivity, as in Section 5.4. The Hessian estimate controls the relative metric correction. Since H^5 embeds into $C^{2,\alpha}$ for $0 < \alpha < 1$ in real dimension four, comparison with

the smooth Calabi–Yau potential, as in Section 5.4, identifies this sum with the normalized potential of the Ricci-flat metric in α_ε. This identifies the same nonlinear mechanism in the two geometries. An explicit elliptic K3 construction additionally requires local Green kernels, their global assembly, and proofs of (7.1)–(7.2); those tasks lie beyond the present Kummer construction.

A The periodic Green kernel and Ewald representation

In this appendix we collect the explicit formulas for the periodic Green kernel used in the exterior problem. We derive an absolutely convergent Ewald representation, justify termwise differentiation, normalize the periodic multipoles at the Kummer points, and record the Fourier-moment formula used in Section 6.

Throughout this appendix, $T = \mathbb{R}^4/\mathbb{Z}^4$ has volume one and $\Delta_0 = \sum_{j=1}^4 \partial_{x_j}^2$. Thus $-\Delta_0$ is nonnegative, and its fundamental solution on $\mathbb{R}^4$ satisfies $-\Delta_0(1/(4\pi^2|x|^2)) = \delta_0$.

A.1 The normalized torus Green function. Let G_T be the unique distribution satisfying $-\Delta_0 G_T = \delta_0 - 1$ and $\int_T G_T\, dx = 0$. Its Fourier expansion is

$$G_T(x) = \sum_{n\in\mathbb{Z}^4\setminus\{0\}} \frac{e^{2\pi\sqrt{-1}n\cdot x}}{4\pi^2|n|^2}, \tag{A.1}$$

where the series is understood distributionally. For pointwise evaluation and differentiation we use the heat kernel. Poisson summation gives its two representations:

$$K_t(x) = \sum_{n\in\mathbb{Z}^4} e^{-4\pi^2|n|^2 t} e^{2\pi\sqrt{-1}n\cdot x} = \frac{1}{(4\pi t)^2} \sum_{m\in\mathbb{Z}^4} \exp\!\left(-\frac{|x+m|^2}{4t}\right). \tag{A.2}$$

Since the zero Fourier mode of K_t is one,

$$G_T(x) = \int_0^\infty \big(K_t(x) - 1\big)\, dt, \qquad x \notin \mathbb{Z}^4. \tag{A.3}$$

Fix an arbitrary parameter $\sigma > 0$.

Proposition A.1 (Ewald representation)**.** *For every $x \in \mathbb{R}^4 \setminus \mathbb{Z}^4$, one has*

$$\begin{aligned} G_T(x) = &\sum_{m\in\mathbb{Z}^4} \frac{\exp\big(-|x+m|^2/(4\sigma)\big)}{4\pi^2|x+m|^2} - \sigma \\ &+ \sum_{n\in\mathbb{Z}^4\setminus\{0\}} \frac{e^{-4\pi^2\sigma|n|^2}}{4\pi^2|n|^2} e^{2\pi\sqrt{-1}n\cdot x}. \end{aligned} \tag{A.4}$$

Both series are absolutely and locally uniformly convergent away from $\mathbb{Z}^4$.

Proof. Split (A.3) at $t = \sigma$. For $r > 0$, direct integration gives $\int_0^\sigma e^{-r^2/(4t)}(4\pi t)^{-2}\, dt = e^{-r^2/(4\sigma)}/(4\pi^2 r^2)$. Using the two representations in (A.2), we obtain

$$\begin{aligned} \int_0^\sigma (K_t(x) - 1)\, dt &= \sum_{m\in\mathbb{Z}^4} \int_0^\sigma \frac{e^{-|x+m|^2/(4t)}}{(4\pi t)^2}\, dt - \sigma \\ &= \sum_{m\in\mathbb{Z}^4} \frac{e^{-|x+m|^2/(4\sigma)}}{4\pi^2|x+m|^2} - \sigma, \\ \int_\sigma^\infty (K_t(x) - 1)\, dt &= \sum_{n\neq 0} \frac{e^{-4\pi^2\sigma|n|^2}}{4\pi^2|n|^2} e^{2\pi\sqrt{-1}n\cdot x}. \end{aligned}$$

Adding the two expressions proves the formula. □

The right-hand side of (A.4) is independent of σ. Thus the Ewald parameter may be chosen solely for computational convenience.

A.2 Differentiation and truncation.

Proposition A.2 (Termwise differentiation)**.** *Let $K \Subset \mathbb{R}^4 \setminus \mathbb{Z}^4$ and let β be a multi-index. Then $\partial^\beta G_T$ is obtained by differentiating both series in* (A.4) *term by term. The differentiated series converge absolutely and uniformly on K.*

Proof. For the reciprocal-lattice series, differentiation introduces only a polynomial factor in $|n|$, while $e^{-4\pi^2\sigma|n|^2}$ provides Gaussian decay. For the real-space series, every derivative of $e^{-|x+m|^2/(4\sigma)}/|x+m|^2$ is a finite sum of polynomial and rational factors multiplied by a Gaussian. Since K stays a positive distance from the lattice, the terms are dominated by $C_{\beta,\sigma}(1+|m|)^{N_\beta}e^{-c_\sigma|m|^2}$. The result follows from the Weierstrass test. □

In particular, if the two Ewald sums are truncated at $|m| \le M$ and $|n| \le N$, then on every compact subset away from the lattice,

$$\left|\partial^\beta G_T - \partial^\beta G_T^{M,N}\right| \le C_\beta \left[(1+M)^{N_\beta}e^{-cM^2/\sigma} + (1+N)^{N_\beta}e^{-c\sigma N^2}\right].$$

Thus every derivative used in the construction has an exponentially convergent representation.

A.3 The regular part at the origin. Set $\mathscr{H}(x) = G_T(x) - 1/(4\pi^2|x|^2)$. This function extends real analytically across the origin. Indeed, the $m = 0$ term in the real-space Ewald sum satisfies

$$\frac{e^{-|x|^2/(4\sigma)} - 1}{4\pi^2|x|^2} = \frac{1}{4\pi^2}\sum_{j=1}^{\infty}\frac{(-1)^j|x|^{2j-2}}{j!(4\sigma)^j},$$

while all remaining terms are analytic near $x = 0$. Its value at the origin is therefore

$$\mathscr{H}(0) = \sum_{m\neq 0}\frac{e^{-|m|^2/(4\sigma)}}{4\pi^2|m|^2} + \sum_{n\neq 0}\frac{e^{-4\pi^2\sigma|n|^2}}{4\pi^2|n|^2} - \sigma - \frac{1}{16\pi^2\sigma}.$$

Away from the origin, $-\Delta_0 G_T = -1$. Since the Euclidean fundamental solution is harmonic there, $\Delta_0\mathscr{H} = 1$. As $\Delta_0|x|^2 = 8$, we may write

$$\mathscr{H}(x) = h_0 + \frac{|x|^2}{8} + \mathscr{H}_{\mathrm{har}}(x), \qquad \mathscr{H}_{\mathrm{har}} = \sum_{\substack{\ell\ge 4\\ \ell \text{ even}}} H_\ell, \quad H_\ell \in \mathcal{H}_\ell^{B_4}.$$

Here $h_0 = \mathscr{H}(0)$, and $\mathscr{H}_{\mathrm{har}}$ is harmonic near the origin and vanishes to order at least four. The expansion follows from invariance of $\mathscr{H}$ under the hyperoctahedral group $B_4 = (\mathbb{Z}_2)^4 \rtimes S_4$ of the square lattice.

A.4 The quotient Green kernel. Let $q : T \to Y = T/\{\pm 1\}$ be the quotient map. For a general source the kernel is $G_Y(x,y) = G_T(\widetilde{x}-\widetilde{y}) + G_T(\widetilde{x}+\widetilde{y})$, as in (2.19). Source derivatives at a fixed point are taken in these covering coordinates. For an orbifold point p_ν, choose a two-torsion lift $\widetilde{p}_\nu \in T$ and define $G_Y(x,p_\nu) = 2G_T(\widetilde{x}-\widetilde{p}_\nu)$. Because G_T is even and $\widetilde{p}_\nu$ is two-torsion, this is independent of the choice of lift. The factor two is forced by $\mathrm{Vol}(Y) = 1/2$, and one has

$$-\Delta_0 G_Y(\,\cdot\,,p_\nu) = \delta_{p_\nu} - 2, \qquad \int_Y G_Y(x,p_\nu)\,dV_0(x) = 0. \tag{A.5}$$

Near p_ν, $G_Y(x,p_\nu) = 1/(2\pi^2 r_\nu^2) + 2\mathscr{H}(x-p_\nu)$. The coefficient $1/(2\pi^2)$ is twice the Euclidean coefficient because the link of an A_1-orbifold point is $S^3/\{\pm 1\}$.

A.5 Periodic multipoles. Let $P_{\ell,\gamma}(x) = r^\ell Y_{\ell,\gamma}(\theta)$ be a homogeneous harmonic polynomial of even degree ℓ. Hobson's formula and the sign from source differentiation give

$$P_{\ell,\gamma}(\partial_x)|x|^{-2} = (-1)^\ell 2^\ell \ell!\, r^{-\ell-2} Y_{\ell,\gamma}(\theta),$$
$$P_{\ell,\gamma}(\partial_y) G_Y(x,y)\big|_{y=p_\nu} = \frac{2^\ell \ell!}{2\pi^2} r_\nu^{-\ell-2} Y_{\ell,\gamma} + O(1).$$

We therefore define the normalized periodic multipole by

$$G_{\nu,\ell,\gamma}(x) = \frac{2\pi^2}{2^\ell \ell!} P_{\ell,\gamma}(\partial_y) G_Y(x,y)\big|_{y=p_\nu}. \tag{A.6}$$

It satisfies $G_{\nu,\ell,\gamma} = r_\nu^{-\ell-2} Y_{\ell,\gamma} + O(1)$ near p_ν. For the positive even degrees used here, differentiation annihilates the constant term in (A.5), so $\Delta_0 G_{\nu,\ell,\gamma} = 0$ away from p_ν.

A.6 The symmetric sixteen-point sum. Let $\mathcal{P} = (\frac{1}{2}\mathbb{Z}/\mathbb{Z})^4$ and define the completely symmetric monopole sum $\mathcal{G}_{\mathrm{K}}(x) = \sum_{p\in\mathcal{P}} G_Y(x,p)$. Since $\sum_{p\in\mathcal{P}} e^{-2\pi\sqrt{-1}n\cdot p}$ equals 16 for $n \in (2\mathbb{Z})^4$ and zero otherwise, one has, in the distributional sense,

$$\mathcal{G}_{\mathrm{K}}(x) = \frac{2}{\pi^2} \sum_{k\in\mathbb{Z}^4\setminus\{0\}} \frac{e^{4\pi\sqrt{-1}k\cdot x}}{|k|^2}.$$

This explains the $4\pi k$ Fourier frequencies appearing in Section 6.

A.7 Fourier moments on a Kummer cell. Let $C = [-1/4, 1/4]^4$, $D_R = C \setminus B_R$, and $0 < R < 1/4$. For a homogeneous harmonic polynomial P_ℓ of degree ℓ, define $\Lambda_{P_\ell}(n) = \int_{D_R} P_\ell(x) e^{-4\pi\sqrt{-1}n\cdot x}\, dx$. Put $\xi = 4\pi n$. The identity $P_\ell(x)e^{-\sqrt{-1}\xi\cdot x} = \sqrt{-1}^\ell P_\ell(\partial_\xi) e^{-\sqrt{-1}\xi\cdot x}$ gives $\Lambda_{P_\ell}(n) = \sqrt{-1}^\ell P_\ell(\partial_\xi)\widehat{\mathbf{1}}_{D_R}(\xi)\Big|_{\xi=4\pi n}$. The cube and ball transforms in four dimensions are

$$\widehat{\mathbf{1}}_C(\xi) = \prod_{j=1}^4 \frac{2\sin(\xi_j/4)}{\xi_j}, \qquad \widehat{\mathbf{1}}_{B_R}(\xi) = 4\pi^2 R^2 \frac{J_2(R|\xi|)}{|\xi|^2}.$$

The apparent singularities at $\xi_j = 0$ and $\xi = 0$ are removable: the corresponding limits are $1/2$ and $\pi^2 R^4/2$, respectively. Consequently,

$$\Lambda_{P_\ell}(n) = \sqrt{-1}^\ell P_\ell(\partial_\xi) \left[\prod_{j=1}^4 \frac{2\sin(\xi_j/4)}{\xi_j} - 4\pi^2 R^2 \frac{J_2(R|\xi|)}{|\xi|^2} \right]\Bigg|_{\xi=4\pi n}. \tag{A.7}$$

For $\ell = 0$, $|D_R| = 1/16 - \pi^2 R^4/2$. If $n \neq 0$, the cube contribution vanishes at $\xi = 4\pi n$, giving $\Lambda_1(n) = -R^2 J_2(4\pi R|n|)/(4|n|^2)$.

A.8 Explicitness of the periodic data.

Proposition A.3 (Explicit periodic data)**.** *Every periodic quantity entering the construction of the exterior Dirichlet and harmonic operators is determined by the following operations:*

1. *the Ewald representation* (A.4)*;*
2. *termwise differentiation justified by Proposition A.2;*
3. *the normalized multipole formula* (A.6)*;*
4. *the Fourier-moment formula* (A.7)*;*
5. *finite-dimensional spherical-harmonic projection.*

Every Ewald truncation has an exponentially decaying remainder.

B Radial Green kernels on the Kummer caps

We construct the separated Green kernels on the sixteen Kummer caps. Each cap metric is radial, agrees with Eguchi–Hanson near its exceptional curve, and is flat in a fixed boundary collar.

We work on one cap and suppress its index ν. We complexify the scalar operator and use the Hermitian angular inner product. Complex conjugation interchanges the (p,q) and (q,p) spaces, while the radial coefficients depend only on $p+q$ and pq. Conjugate modes therefore give real solutions for real sources and boundary values.

B.1 Radial Kähler metrics. For a radial potential $\Phi(\rho)$, where $\rho = |z_1|^2 + |z_2|^2$, set $A = \Phi'$ and $B = \Phi' + \rho\Phi''$. The metric and its inverse are

$$\begin{aligned} g_{i\bar{j}} &= A\delta_{ij} + A'\bar{z}_i z_j, \\ g^{i\bar{j}} &= \frac{1}{A}\delta_{ij} - \frac{A'}{AB} z_i \bar{z}_j. \end{aligned} \tag{B.1}$$

Its volume form is $\omega^2/2 = 4AB\, dV_{\mathbb{R}^4}$. For the background of Section 2.4,

$$A = A_a^{\mathrm{mod}} = \frac{\sqrt{D_a(\rho)}}{2\rho}, \qquad B = B_a^{\mathrm{mod}} = \frac{\rho(1-\beta_a(\rho))}{2\sqrt{D_a(\rho)}}, \qquad 4AB = 1-\beta_a.$$

Near the exceptional divisor these reduce to

$$A = \frac{\sqrt{\rho^2+a^4}}{2\rho}, \qquad B = \frac{\rho}{2\sqrt{\rho^2+a^4}}.$$

On the fixed outer collar $r_4 \le r \le R$, one has $A = B = 1/2$.

B.2 Bihomogeneous harmonic modes. Let $P_{p,q,\gamma}$ be a bihomogeneous Euclidean harmonic polynomial of bidegree (p,q), with p,q nonnegative integers. With $\ell = p+q$, write $P_{p,q,\gamma}(r\theta) = r^\ell Y_{p,q,\gamma}(\theta)$. Only even ℓ descend to the Kummer quotient. Bihomogeneity gives

$$\sum_j z_j \partial_{z_j} P_{p,q,\gamma} = pP_{p,q,\gamma}, \qquad \sum_j \bar{z}_j \partial_{\bar{z}_j} P_{p,q,\gamma} = qP_{p,q,\gamma}, \tag{B.2}$$

and harmonicity gives $\sum_j \partial_{z_j}\partial_{\bar{z}_j} P_{p,q,\gamma} = 0$. We seek solutions of the form $u = w(\rho)P_{p,q,\gamma}$.

Proposition B.1 (Separated radial equation)**.** *One has*

$$\Lambda_\omega \sqrt{-1}\partial\bar{\partial} u = \frac{\rho A w'' + \big((\ell+2)A + \rho A'\big)w' - pqA'w}{AB} P_{p,q,\gamma}. \tag{B.3}$$

Thus $-\Lambda_\omega\sqrt{-1}\partial\bar{\partial}(wP_{p,q,\gamma}) = F(\rho)P_{p,q,\gamma}$ *is equivalent to*

$$-\Big(\rho^{\ell+2}Aw'\Big)' + pq\,\rho^{\ell+1}A'w = \rho^{\ell+1}AB\,F. \tag{B.4}$$

Proof. Writing $P = P_{p,q,\gamma}$, differentiation gives

$$u_{i\bar{j}} = w''\bar{z}_i z_j P + w'\big(\delta_{ij}P + \bar{z}_i P_{\bar{j}} + z_j P_i\big) + wP_{i\bar{j}}.$$

Contracting with (B.1) and using (B.2) gives (B.3). Multiplication by $\rho^{\ell+1}AB$ then gives (B.4). □

Set $\mathsf{p}_{\ell,a} = \rho^{\ell+2}A$ and $\mathsf{q}_{p,q,a} = pq\,\rho^{\ell+1}A'$. The homogeneous equation is

$$-(\mathsf{p}_{\ell,a}w')' + \mathsf{q}_{p,q,a}w = 0. \tag{B.5}$$

B.3 The regular solution at the exceptional divisor. As $\rho \to 0$, $A(\rho) = \frac{a^2}{2\rho}(1 + O(\rho^2/a^4))$. The indicial equation of (B.5) is $s(s+\ell)+pq = (s+p)(s+q) = 0$, with roots $-p$ and $-q$. Set $m = \min\{p,q\}$ and $M = \max\{p,q\}$.

Lemma B.2 (Regular radial solution)**.** *Up to a multiplicative constant, there is a unique solution $w^{\mathrm{reg}}_{p,q,a}$ of* (B.5) *for which $w^{\mathrm{reg}}_{p,q,a}P_{p,q,\gamma}$ extends smoothly across the exceptional divisor. Normalize it by $w^{\mathrm{reg}}_{p,q,a}(\rho) = \rho^{-m}(1+O(\rho^2/a^4))$ as $\rho \to 0$. In the exact Eguchi–Hanson region it has the Frobenius expansion*

$$w^{\mathrm{reg}}_{p,q,a}(\rho) = \rho^{-m}\sum_{j=0}^{\infty} c_j^{p,q}\left(\frac{\rho^2}{a^4}\right)^j, \qquad c_0^{p,q} = 1, \tag{B.6}$$

convergent for $0 < \rho < \min\{a^2, r_1^2\}$, with coefficients

$$c_j^{p,q} = -\frac{(2j-m-2)(2j+M-1)}{2j(2j+M-m)}c_{j-1}^{p,q}, \qquad j \geq 1. \tag{B.7}$$

Proof. The origin is a regular singular point of (B.5). Its indicial roots are $-p$ and $-q$. In a resolution chart, the branch corresponding to $-m$ extends smoothly across the exceptional divisor. The other branch is singular; when $p = q$, the second independent solution has a logarithmic singularity.

In the exact Eguchi–Hanson region the equation becomes

$$\rho^2(\rho^2+a^4)w'' + \rho\big((\ell+2)\rho^2 + (\ell+1)a^4\big)w' + pq\,a^4 w = 0.$$

Substituting (B.6) gives (B.7), whose denominator is nonzero for $j \geq 1$. The ratio test gives convergence for $\rho < a^2$, with possible termination; the background uses this equation only for $\rho < r_1^2$.

In a resolution chart (ζ, t), where $\rho = |t|(1+|\zeta|^2)$, each monomial of $\rho^{-m}P_{p,q,\gamma}$ is $t^{(p-q)/2}$, when $p \geq q$, or its conjugate when $q \geq p$, times a smooth function of ζ. These exponents are integers because ℓ is even, and every additional factor ρ^{2j} is smooth. The other branch has a negative fiber power, or a logarithm when $p = q$, and cannot extend smoothly. □

Continue the regular solution through the fixed transition annulus using (B.5). This continuation also has a convergent iterated-integral representation. Choose $0 < \rho_* < \min\{a^2, r_1^2\}$ and set

$$V = \begin{pmatrix} w \\ \mathsf{p}_{\ell,a}w' \end{pmatrix}, \qquad \mathsf{C}(\rho) = \begin{pmatrix} 0 & \mathsf{p}_{\ell,a}^{-1} \\ \mathsf{q}_{p,q,a} & 0 \end{pmatrix}.$$

Then $V' = \mathsf{C}V$, and for $\rho_* \leq \rho \leq R^2$,

$$V(\rho) = \left[I + \sum_{j=1}^{\infty}\int_{\rho_* < s_j < \cdots < s_1 < \rho} \mathsf{C}(s_1)\cdots\mathsf{C}(s_j)\,ds_j\cdots ds_1\right]V(\rho_*). \tag{B.8}$$

With $L_{p,q,a} = \int_{\rho_*}^{R^2}\|\mathsf{C}(s)\|\,ds$, the tail after order J is bounded by

$$\|V(\rho_*)\|e^{L_{p,q,a}}\frac{L_{p,q,a}^{J+1}}{(J+1)!}.$$

This bound applies to each fixed a, p, q; it is not uniform in the angular modes.

Lemma B.3 (Nonvanishing)**.** *The regular solution satisfies $w^{\mathrm{reg}}_{p,q,a}(\rho) \neq 0$ for $0 < \rho \leq R^2$.*

Proof. If $w^{\mathrm{reg}}_{p,q,a}(\rho_0) = 0$ for some $\rho_0 > 0$, then $w^{\mathrm{reg}}_{p,q,a}P_{p,q,\gamma}$ would be a nonzero smooth harmonic function on the resolved region bounded by $r^2 = \rho_0$, with zero Dirichlet boundary value. The Dirichlet energy identity forces this function to vanish, a contradiction. □

B.4 The zero-boundary second solution. Reduction of order gives

$$w^D_{p,q,a}(\rho) = w^{\mathrm{reg}}_{p,q,a}(\rho) \int_\rho^{R^2} \frac{ds}{s^{\ell+2}A(s)\big(w^{\mathrm{reg}}_{p,q,a}(s)\big)^2}. \tag{B.9}$$

This solution vanishes at R^2 and has the normalized Wronskian

$$\rho^{\ell+2}A(\rho)\left[w^D_{p,q,a}(w^{\mathrm{reg}}_{p,q,a})' - (w^D_{p,q,a})' w^{\mathrm{reg}}_{p,q,a}\right] = 1.$$

B.5 The radial Dirichlet Green kernel. For $\rho_< = \min\{\rho,\sigma\}$ and $\rho_> = \max\{\rho,\sigma\}$, define

$$g^D_{p,q,a}(\rho,\sigma) = w^{\mathrm{reg}}_{p,q,a}(\rho_<) w^D_{p,q,a}(\rho_>). \tag{B.10}$$

Proposition B.4 (Radial Green formula)**.** *Suppose that $F(\rho)P_{p,q,\gamma}$ extends smoothly over the resolved cap, including its exceptional divisor and boundary. The unique smooth solution $wP_{p,q,\gamma}$ of $-\Lambda_{\omega_a^{\mathrm{mod}}}\sqrt{-1}\partial\bar\partial(wP_{p,q,\gamma}) = F(\rho)P_{p,q,\gamma}$, with zero boundary value at $r = R$, is given by*

$$w(\rho) = \int_0^{R^2} g^D_{p,q,a}(\rho,\sigma)\sigma^{\ell+1}A(\sigma)B(\sigma)F(\sigma)\,d\sigma. \tag{B.11}$$

Proof. Away from $\rho = \sigma$, the kernel solves the homogeneous equation. The Wronskian identity gives

$$\partial_\rho g(\sigma^+,\sigma) - \partial_\rho g(\sigma^-,\sigma) = -\frac{1}{\sigma^{\ell+2}A(\sigma)}.$$

Thus $-[\sigma^{\ell+2}A(\sigma)\partial_\rho g]^{\sigma^+}_{\sigma^-} = 1$, as required by (B.4). The zero boundary condition follows from $w^D(R^2) = 0$. For the stated source, the corresponding angular component of the smooth Dirichlet solution on the resolved cap satisfies this equation and the regular endpoint condition. Variation of parameters with the normalized Wronskian gives (B.11), and Dirichlet uniqueness identifies the expression with that smooth solution. □

B.6 The full cap Dirichlet inverse. Let $L = S^3/\{\pm1\}$, and choose a complex orthonormal basis $Y_{p,q,\gamma}$ for its standard quotient measure $d\sigma_L$, so that $\int_L Y_I\overline{Y_J}\,d\sigma_L = \delta_{IJ}$. Expand a source as $f(r,\theta) = \sum_{p,q,\gamma} f_{p,q,\gamma}(r)Y_{p,q,\gamma}(\theta)$. Since $P_{p,q,\gamma} = \rho^{\ell/2}Y_{p,q,\gamma}$, its coefficient in the polynomial normalization is $F_{p,q,\gamma}(\rho) = \rho^{-\ell/2}f_{p,q,\gamma}(\sqrt{\rho})$. The cap inverse is therefore

$$\begin{aligned} D_{\mathrm{cap},a}f &= \sum_{p,q,\gamma} w_{p,q,\gamma}(\rho)P_{p,q,\gamma}, \\ w_{p,q,\gamma}(\rho) &= \int_0^{R^2} g^D_{p,q,a}(\rho,\sigma)\sigma^{\ell+1}A(\sigma)B(\sigma)F_{p,q,\gamma}(\sigma)\,d\sigma. \end{aligned} \tag{B.12}$$

The volume measure is $dV_a = 2\rho A(\rho)B(\rho)\,d\rho\,d\sigma_L$, so the full kernel is

$$G^D_{\mathrm{cap},a}(x,y) = \frac{1}{2}\sum_{p,q,\gamma}(\rho_x\rho_y)^{\ell/2} g^D_{p,q,a}(\rho_x,\rho_y)Y_{p,q,\gamma}(\theta_x)\overline{Y_{p,q,\gamma}(\theta_y)}.$$

This formula is initially interpreted distributionally. For $\rho_x \neq \rho_y$, radial separation gives locally normal convergence, including differentiated series away from the endpoints. At equal radii, use Abel regularization by inserting $e^{-t\ell(\ell+2)}$ and letting $t \downarrow 0$; the regularized kernel converges to the Green kernel off the diagonal. Absolute convergence at equal radii does not follow from $x \neq y$ alone. After integration against a smooth source, angular synthesis gives the smooth Dirichlet solution, with regularity at the exceptional divisor imposed by the regular branch. Equivalently, these interpretations follow by expanding the unique fixed-a Dirichlet solution into compact $U(2)$ angular representations and applying the radial formula.

B.7 Harmonic extension and the Dirichlet-to-Neumann map. The regular harmonic extension of a boundary mode is

$$H_{\mathrm{cap},a}Y_{p,q,\gamma} = \frac{r^\ell w^{\mathrm{reg}}_{p,q,a}(r^2)}{R^\ell w^{\mathrm{reg}}_{p,q,a}(R^2)} Y_{p,q,\gamma}. \tag{B.13}$$

The metric is flat near $r = R$, where the outward unit normal is ∂_r. Its Dirichlet-to-Neumann eigenvalue is therefore

$$\lambda^{\mathrm{cap}}_{p,q}(a) = \frac{\ell}{R} + 2R\frac{(w^{\mathrm{reg}}_{p,q,a})'(R^2)}{w^{\mathrm{reg}}_{p,q,a}(R^2)}. \tag{B.14}$$

Equivalently, with $\mathcal{U}_{p,q,a}(r) = r^\ell w^{\mathrm{reg}}_{p,q,a}(r^2)$, this is $\partial_r \log \mathcal{U}_{p,q,a}(r)|_{r=R}$.

B.8 The fixed flat collar. On $r_4 \le r \le R$, the homogeneous equation reduces to $\rho w'' + (\ell+2)w' = 0$, so

$$w(\rho) = c_+ + c_-\rho^{-\ell-1}, \qquad \mathcal{U}(r) = c_+ r^\ell + c_- r^{-\ell-2}.$$

With the dimensionless reflection coefficient $\mathfrak{r}_{p,q}(a) = (c_-/c_+)R^{-2\ell-2}$, the boundary coefficient becomes

$$\lambda^{\mathrm{cap}}_{p,q}(a) = \frac{1}{R}\frac{\ell - (\ell+2)\mathfrak{r}_{p,q}(a)}{1+\mathfrak{r}_{p,q}(a)}. \tag{B.15}$$

The interior Dirichlet-to-Neumann coefficient λ_4 at r_4 is nonnegative by the energy identity. Collar matching gives

$$\frac{c_-}{c_+} r_4^{-2\ell-2} = \frac{\ell - r_4\lambda_4}{\ell + 2 + r_4\lambda_4}.$$

In particular, $c_+ \neq 0$ and $|\mathfrak{r}_{p,q}(a)| \le (r_4/R)^{2\ell+2}$. This bound gives the uniform smoothing used in Section 4.5 and controls the denominator in (B.15).

B.9 The flat-orbifold limit. We record the fixed-mode geometric limit of the cap data. The uniform inverse estimate in Section 4.6 uses exterior coercivity and does not require this limiting argument.

Proposition B.5 (Fixed-mode flat limit)**.** *For every fixed p, q, γ, $\lambda^{\mathrm{cap}}_{p,q}(a) \to \ell/R$ as $a \to 0$, or, equivalently, $\mathfrak{r}_{p,q}(a) \to 0$.*

Proof. Let u_a be the harmonic extension of $Y_{p,q,\gamma}$ from $r = R$. The maximum principle, applied to the subharmonic function $|u_a|^2$, gives a uniform supremum bound from this fixed boundary value. Away from the exceptional point the metrics converge smoothly to the flat orbifold metric. Local elliptic estimates therefore give subsequential smooth convergence $u_a \to u_0$ on compact subsets of the punctured ball. On the fixed flat boundary collar, boundary Schauder estimates also give convergence up to $r = R$, so the limit has boundary value $Y_{p,q,\gamma}$.

The limit is bounded, so its lift to the double cover has a removable harmonic singularity at the origin. It is consequently the regular flat harmonic extension

$$u_0(r,\theta) = (r/R)^\ell Y_{p,q,\gamma}(\theta).$$

This identifies the limit independently of the subsequence. The same boundary estimates give convergence of the normal derivatives at $r = R$, proving the asserted limit. □

B.10 Summary of the cap construction.

Proposition B.6 (Explicit cap data)**.** *For sufficiently small $a > 0$, the cap operators used in Section 3 are determined by the regular radial solutions, obtained from the convergent Frobenius expansion and one-dimensional continuation, and the zero-boundary solutions* (B.9)*. These give the radial Green kernels* (B.10)*, the Dirichlet inverse* (B.12)*, the harmonic extension* (B.13)*, and the Dirichlet-to-Neumann coefficients* (B.14)*. After angular projection, their construction uses one-dimensional differentiation and integration, convergent radial series, and angular synthesis.*

C Angular bases and Fourier moments

The following optional algebraic reductions assist evaluation of the representation. They are not needed for the Green estimate or the convergence theorem. We give the symmetry reduction, harmonic projection, measure normalization, and polynomial–Fourier moments together.

C.1 Symmetries preserving the complex structure. The holomorphic unitary lattice group used here is

$$\Gamma = \{(z_1, z_2) \mapsto (\zeta_1 z_{\sigma(1)}, \zeta_2 z_{\sigma(2)}) : \zeta_j \in \{1, -1, \sqrt{-1}, -\sqrt{-1}\},\ \sigma \in S_2\}.$$

Its action lifts to the resolution and preserves the radial background, its volume form, and the equation. Translation by each two-torsion point also lifts and permutes the sixteen identical caps. Uniqueness of the mean-zero inverse, or induction on the tree recursion, shows that u_a has these symmetries. The larger real signed-permutation group B_4 preserves the flat torus Green kernel, but its arbitrary real-coordinate permutations need not preserve the complex structure or the Eguchi–Hanson metric. It cannot be used to discard modes in the full cap or nonlinear problem. The Green operator for arbitrary sources uses the entire even harmonic basis. For the Γ-invariant sources arising in the correction recursion, one may instead use the invariant basis constructed here.

C.2 Exact harmonic projection. Let P be a homogeneous polynomial of degree ℓ on $\mathbb{R}^4$. Its harmonic projection is

$$\Pi_\ell P = \sum_{j=0}^{\lfloor \ell/2 \rfloor} (-1)^j \frac{(\ell - j)!}{4^j j!\, \ell!} |x|^{2j} \Delta_0^j P.$$

Thus harmonic projection is a finite algebraic operation.

For a real homogeneous polynomial P, form the finite group average $\mathcal{R}_\Gamma P(x) = |\Gamma|^{-1} \sum_{g \in \Gamma} P(gx)$. Because Γ is orthogonal, harmonic projection commutes with this average. A spanning family for its invariant harmonics of degree ℓ is therefore $\{\Pi_\ell \mathcal{R}_\Gamma x^\nu : |\nu| = \ell\}$. Remove linear dependencies by exact row reduction before forming the Gram matrix. All coefficients are rational. The absence of invariant harmonic quadratics follows because independent phase rotations eliminate the non-Hermitian quadratic terms, while coordinate interchange leaves only a multiple of $|z_1|^2 + |z_2|^2$, which is not harmonic. This optional symmetry reduction is not used in the Green estimate or the convergence proof.

C.3 Exact spherical Gram matrices. For stable reconstruction it is convenient to orthogonalize the canonical harmonic basis exactly.

Normalize

$$\langle f, g \rangle_S = \frac{1}{2\pi^2} \int_{S^3} f g \, d\sigma. \tag{C.1}$$

If $a_1, \dots, a_4$ are nonnegative integers and $A = a_1 + \cdots + a_4$, then

$$\frac{1}{2\pi^2} \int_{S^3} x_1^{2a_1} \cdots x_4^{2a_4} \, d\sigma = \frac{1}{(A+1)!} \prod_{j=1}^{4} \frac{(2a_j)!}{4^{a_j} a_j!}.$$

Monomials with an odd exponent have zero integral. Hence every Gram matrix $G_\ell = (\langle H_I, H_J \rangle_S)_{I,J}$ has rational entries.

Choose an exact factorization $G_\ell = L_\ell D_\ell L_\ell^T$. Then $E_\ell = D_\ell^{-1/2} L_\ell^{-1} H_\ell$ is orthonormal for (C.1). For the standard quotient measure on $S^3/\{\pm 1\}$, whose total volume is π^2, the corresponding basis is $Y_I = E_I/\pi$. For the physical boundary measure on Σ_ν, the basis becomes $R^{-3/2} Y_I$. These rescalings reconcile the normalized Gram calculation with the angular kernels and the matrix (3.12). To apply the separated cap formulas, decompose this real basis into the complex (p, q) angular spaces.

This avoids the artificial loss which may arise from reconstructing coefficients in a poorly conditioned monomial basis.

C.4 Fourier moments. The fully Kummer-symmetric Fourier expansion uses frequencies $4\pi n$, where $n \in \mathbb{Z}^4$. Let $C = [-1/4, 1/4]^4$ and $D_R = C \setminus B_R$. For a homogeneous harmonic polynomial H of degree ℓ, put $\Lambda_H(n) = \int_{D_R} H(x) e^{-4\pi\sqrt{-1} n \cdot x} \, dx$. Formula (A.7), with $P_\ell = H$, evaluates these polynomial–Fourier volume moments by differentiation of the explicit elementary–Bessel expression at $\xi = 4\pi n$. Boundary matrix elements are computed from the separate periodic trace and normal-derivative formulas.

On the flat exterior, the Fourier coefficients of a nonlinear source may first be assembled by convolution and then contracted against all required harmonic moments. The expensive Fourier convolution need not be repeated separately for each harmonic channel.

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